



Fundamentals of Computing

Sciences

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ComSci Dep't, College of Arts &

Mindanao State University -
Maguindanao

FUNDAMENTALS OF COMPUTING

Course Code: **CCC100**

Overview:

This course provides an overview of the Computing Industry and Computing profession, including research and applications in different fields; an appreciation of computing in different fields such as Biology, Sociology, Environment and Gaming; and understanding of ACM Requirements; an appreciation of the history of computing; and knowledge of the key components of computer systems (Organization and Architecture), Malware, Computer Security, Internet and Internet protocols, HTML 4/5 and CSS.

General Learning Outcomes:

At the end of the semester, the students can:

1. Explain the fundamental principles, concepts and evolution of computing systems as they relate to different fields.
2. Expound on the recent developments in the different computing knowledge areas.
3. Analyze solutions employed by organizations to address different computing issues.

Duration: 30 hours at 2 hours per week lecture

Contents / Topics:

1. Vision, Mission, Goal, Core Values and Objectives of the University, College and Department1
2. Industry in the Profession
3. Appreciation of Computing in Different Fields.....
4. Different Specializations
5. Evolution of Computing
6. Key Components of a Computer System
7. Key Components of Operating System
8. Malware
9. Complex Security
10. Network, Internet, and Internet Protocols
11. HTML and CSS
12. Computer Systems (Organization and Architecture)

Methodology:

1. Oral Presentation

2. Assignments
3. Quizzes
4. Major Exams

Evaluation Criteria:

Prelim Exam	20%
Midterm Exam	20%
Final Exam.....	20%
Performance Task	25%
Written Task/Quiz	15%
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	100%

References/Textbook/Suggested Readings

Cashman, S. & Vermaat, M.E. (2014) Discovering Computers. Cengage Learning

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LESSON 1: VISION, MISSION, CORE VALUES, GOALS AND OBJECTIVES

OVERVIEW:

This lesson introduces the Vision Mission, Goals of the University, College and the Department in charge of the offering of the course.

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. describe the University's Vision, Mission, Goals and its philosophy;
2. express the College's Vision, Mission, Goals and its philosophy; and
3. explain the Department's Vision, Mission, Goals and its philosophy.

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

90 minutes

LESSON PROPER:

MINDANAO STATE UNIVERSITY-MAGUINDANAO

VISION

An Educational Zone of Peace, offering world class higher learning in Maguindanao for Bangsamoro's continuing growth for all.

MISSION

Driven by the commitment to develop highly competent leaders and workers in agriculture, fisheries, forestry, and other fields: offers research, extension, and production services for the development of Bangsamoro.

GOALS

1. Graduate proficient professionals and globally ready citizens;
2. Foster strong faculty research culture and tradition;
3. Expand outreach and deepen impact of extension initiatives with communities;
4. Level up production toward practice- oriented students' learning and budget augmentation;
5. Provide an enabling environment conducive to operational excellence; and

6. Enhance MSU-Maguindanao organizational learning and innovating capacity.

OBJECTIVES

1. Deliver high quality instructional standards;
2. Internationalize students' perspective and experiences;
3. Strengthen faculty qualifications;
4. Affirm faculty members scholarly feats through research;
5. Incentivize faculty members' conduct of research;
6. Rationalize/intensify University's extension policy/ program;
7. Rationalize/intensify university's policy/program;
8. Achieve administrative efficiency;
9. Achieve financial stability; and
10. Capacitate the university in knowledge management competencies.

COLLEGE OF ARTS & SCIENCES

VISION

A highly competitive service College in Maguindanao and its environs.

MISSION

Committed to develop holistic manpower excelling in Computer Technology, Islamic Studies and General Education; engage in research, extension and production for sustainable Maguindanao.

CORE VALUES

Diversity Integrity Respect Excellence Commitment Teamwork Service

GOALS

1. Prepare students mental, physical & emotional skills in the chosen field of study
2. Enhance manpower skills in Information Technology & Islamic Studies
3. Develop responsible leaders in the field of Information Technology & Islamic Studies
4. Strengthen linkages, networks and partnerships among government and private organization at the Local, National and International levels
5. Promote utilization of research and deliver extension services.

COMPUTER SCIENCE DEPARTMENT

VISION

An information hub offering competencies in Computer Technology education to empower Maguindanao and its environs.

MISSION

Committed to upgrade manpower adept with modern technology; get involved in research, extension and production for sustainable Maguindanao.

GOALS

1. Equip learners with essential competencies (knowledge, skills, and values) in the field of Information Technology.
2. Design an enabling environment conducive to the development of learners.
3. Strengthen linkages, network and partnership among public and private organizations in the Local, National, and International levels.
4. Advocate research, engage in extension and development services.

OBJECTIVES

1. Produce effective and quality product of ICT standard.
2. Perform software engineering approaches in creation of system projects.
3. Apply ethical decisions in the implementation of Information Technology practices.
4. Setup resources to emulate real-time technology.
5. Engage in information interchange among agencies and institutions.
6. Conduct researches and extension services

ACTIVITY:

Lecture, Group Discussion

EVALUATION:

1. What are the Core Values of the University and of the College and how will these accomplish the stated the Mission?

LESSON 2: INDUSTRY IN THE PROFESSION

OVERVIEW:

This lesson discusses the different fields of Computing Technology.

LEARNING OUTCOME:

At the end of the lesson, the student can identify the different IT related industries.

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

ACTIVITY:

Lecture, Group Discussions

EVALUATION:

LESSON 3: Appreciation of Computing in Different Fields

OVERVIEW:

This lesson discusses the different fields where Computing Technology is used.

LEARNING OUTCOME:

At the end of the semester, the student can recognize the computer technologies in the different fields.

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

ICT in the Context of Global Communication

ICT has become the main method of communication, getting information and education, attaining services from businesses, hospitals, utilities, government, purchasing products and services, doing personal and business transactions, and expressing social advocacies and awareness.

Education

- Modern homes have computers for research and learning by students and the family for school lessons. They are also used to learn about other people, place, current events, and recent technologies.
- Computers in school are used for fast, easy and effective instruction and learning in Computer, Math, Science, and English. Teachers use images, videos, and Computer-Based Training (CBT) programs to present topics visually to enhance learning.
- Courses on the Internet called Online Instructions are available to learn various lessons such as origami, cooking, and even earn a college degree.
- Libraries are now equipped with computers to organize books and to expand references available to the students through the internet.
- Students use computers to type reports, do their assignments, and communicate with teachers.

- School use computers to keep the database of information about the school and its students.

Telecommunications

- Computers in homes, schools, private and government offices, hospitals, and almost everywhere are used to get in touch with other people.
- Everyone uses the computer to send messages and other information, all over the world. Computers are used to send letters, pictures, video clips, and music. Talking to and seeing friends, relatives and acquaintances are made easier and a lot cheaper by the use of computers connected to the internet.
- Computers are also used in the mobile phone communications to control calls and SMS (short message service) or text traffic.
- VoIP (Voice over Internet Protocol) is a voice communication application via the internet, which costs less than traditional means.

Commerce

- Shops and supermarkets use software which calculates the bills and efficiently processes both cash and credit transactions. Online shopping continues to grow in popularity since it provides a lot of choices for items and pricing from various sources, as well as the convenience of doing everything at home or place where you would rather be.

Sports and Recreation

- There are simulation software where athletes can practice their skills, and the computer will identify the flaws in their techniques.
- Use of computerized fitness programs, gadgets, and equipment
- Computers in television stations and movie outfits are used to by Movie Directors to create special effects for TV and movies, to make watching movies exciting and truly entertaining.
- Computers at home are used to play games, listen to music, and watch movies.

Arts, Design and Music

- Image and graphics design, photo editing, and music arrangements
- Music and videos are shared online by their creators and can be downloaded for free or for a fee.

Media and Visual Arts

- Architecture, interior design, and landscape architecture
- Animation, web design and mobile designs
- Photography, film making, audio and video
- Computers are used by animators to create cartoon characters in the movies

- Characters created by computers look real – for example, dinosaurs look as if they still exist today. Dangerous and almost impossible scenes are now possible with state-of-the-art computer programs run by well-trained artists.

Health Services

- Computers in hospitals are used by doctors to store medical data, patient's records, research materials and other information to effectively diagnose and treat patients.
- Online health information and consultations are now available on the internet. Sharing of valuable experience and expertise helps treat patients with rare and serious diseases faster.
- Computers in operating rooms assist doctors perform microsurgery with accuracy and better chances of success.
- Computers are used in hospitals to monitor the patient's condition. These computers send out alarm signal whenever the patient's condition becomes critical, so doctors and nurses can attend to them immediately.
- Emails are used extensively to communicate with the doctors efficiently. Test results and diagnoses are sent via the internet.
- With the internet, doctors share experiences and medical data with other doctors around the globe to research and diagnose patients with greater accuracy.

Science and Research

- Modern laboratories have computers used by chemist to invent materials, medicine, and products to improve life.
- Physics look for new renewable, safe, and affordable power sources using computers.
- Mathematicians solve complex problems with the use of super-fast computers
- Inventors design and test their products in the computer using simulations to ensure a working model before actual manufacturing.

Travel and Hotel Bookings

- Airline fares and tickets are now paid online; hotel and housing are also reserved using the internet, and even ship travels are also booked online.
- Travel routes and maps can be downloaded to your PDAs or print maps to ensure you will travel faster and not get lost on your trip
- Computers in airports are used by air traffic controllers to control and schedule take-off and landing of planes, to ensure safety.
- Computers on airplanes help both military and commercial pilots to fly and land their airplane safely and efficiently. They use the computer to know how far they are from their destination, how high they are, learn the plane's condition, and many other information.

- Cars, boats, and other gadgets are equipped with navigational aids designed to help the travelers reach their destination faster.
- You can book an air ticket or railway tickets and even make hotel reservations online. Travelers can also browse images of the places they want to see, and even view the amenities inside the hotel in advance.

Business and Accounting

- Computers in groceries are used by cashiers to store prices of goods, determine inventory of goods, and locate the goods in the store.
- Computers in offices are used to store daily business deals and transactions.
- Secretaries type their letters on the computer, print and store them for future use.
- One can always use the computers in the office to do several business activities like planning and scheduling. There are software used to store contact information, generate plans, schedule appointments.
- Call centers use computers to record received or transmitted large volume of requests by telephone to administer incoming product support or information inquiries from consumers or telemarketing.
- Data encoders use computers to enter or update data into a computer system database.
- Medical transcriptionist use computer to convert voice-recorded reports, as dictated by physicians or other healthcare professionals, into text format
- Accountants use computers to know the financial status of the business.
- Accounting process is done using computers, like preparation of financial reports in fast and accurate manner. With gathered and research information, computers can be used to predict future trends of business. Timely and accurate information to business managers will give them better decisions as they would like to direct their business.

Manufacturing Industries

- Automobiles and other vehicles are manufactured using computer-aided machineries and robots; this is called computer-aided manufacturing (CAM). CAM is a programming tool that helps produce better products faster and cheaper with precision and accuracy.
- Computers in manufacturing also manage raw materials, suppliers, cost and inventory
- Computers in industry are also used to design new and better machines, test them in the computer before they are manufactured.
- Some industry are fully automated, meaning all the processes are done by robots under computer control such as in car manufacturing, book publishing, and in some electronic fabrication. This is called Computer Integrated Manufacturing (CIM).

Banking

- Almost all of our financial transactions are done by computer software. They provide ease, security, speed and convenience. Banks usually provide remote services to their clients by installing Automated Teller Machines (ATM) controlled by a network computer, which dispenses cash and client information.
- Banks use computers so that people can get and deposit money any time of the day. They also use computers to record customer's account information. Computers help banks do business away from their location, example of which is the ATM.

Mathematics

- Computerized calculations in the creating of buildings, machines, bridges, airplane and radio communications.
- Solve differential equations and advance numerical algorithms to solve the equations of high performance computers.
- Software can help simulate data and demonstrate in using statistical graphs, like a bar chart, table or diagram mode.

Engineering and Architecture

- Architects design houses, buildings and bridges using Computer Aided Design (CAD) programs which is a lot faster and presentable.
- Civil Engineers use computers to calculate the strength of footing and foundations of a building.
- Electrical engineers design and determine electrical power requirements of buildings and houses.
- Mechanical Engineers use computer in their work to design and build machineries and vehicles such as cars, boats and airplanes with better aesthetics, accuracy, and safety.

Agriculture

- Computers are used in research, monitoring to improve agriculture, crop, and animal production, care and management.
- Food processing and packaging.

Writing and Publishing

- Books, magazines, newspapers, music, and videos are important products to share. Current information, inspiring and creative stories, great computations, entertaining movies, educational books, and a lot more are all products of publishing.

- Print publishing such as books, newspapers, and magazines are designed and prepared using powerful desktop computers using drawing, photo editing, and lay-out programs.
- Videos are edited in powerful and faster computers since videos have higher memory requirements. Video editing programs together with other programs used in print publishing maybe employed together to craft and finish a movie.
- Books are shared online by their creators and can be downloaded for free, or for a fee.
- Read news using online newspapers, read books using ebooks, research for materials and journal online.

Government Service

- Computers in government offices are used by Bureau of Internal Revenue (BIR) to compute taxes of real properties such as homes and buildings.
- The city government uses computers to render service to its citizens; for example, the traffic lights are controlled by computers to lessen congestion of vehicles.
- The National Statistics Office (NSO) uses computers to record the births, marriages, and deaths of the country's citizens. It is also used to record other valuable personal information.
- The Department of Foreign Affairs (DFA) records all citizens traveling abroad. DFA issues computer-generated passports, the latest passports have bar codes included for faster information retrieval and monitoring

Military Applications

- Computers are used to guide weapon systems from any source and pinpoint targets with precision and accuracy.
- Computers on board mechanical vehicles are used to locate and diffuse bombs
- Computers control unmanned airplanes, or drones to locate enemy locations, and deliver firepower when necessary
- Computers use software embedded in many modern weapons to ensure accuracy and precision. Computers also process valuable information or intelligence especially impending terrorist attacks.

Weather Analysis

- There are supercomputers used to analyze and predict the weather of the day. Data gathered from weather satellites will be calculated by these computers to give the wind direction, wind velocity, volume of rainfall, and other important weather information.

Space Exploration

- The National Aeronautics and Space Administration (NASA) in the USA uses computers in spacecraft to explore planets and other heavenly bodies.
- Computers likewise used in Earth control stations for the spacecraft and satellite on mission.
- Computers are also used on satellites and control stations to monitor weather, transmit voice and data, and other applications around the world.
- Computers on board spacecraft are equipped with camera, video, and other sensors that send vital data back to earth.

Aeronautics

- Pilots are trained using software which simulates flying. Flight simulations are safe and cost-effective means of training future pilots without the risk of a real airplane crash. They employ software to enhance both visual and mechanical reflexes to new and even seasoned pilots.

ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

LESSON 4: DIFFERENT SPECIALIZATIONS

OVERVIEW:

This lesson presents the different careers of people engaged in Computer Technology.

LEARNING OUTCOME:

At the end of the lesson, the student can

1. distinguish different IT Job titles and descriptions,
2. Identify the people behind the development of computers

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

Peopleware

- It refers to anything that has to do with the role of people in the development or use of computer software and hardware systems.
- It refers to the user.
- It is the skilled workers in the Information Technology field.

IT GROUP	JOB TITLE
Management	Computer Systems Manager
Systems and Procedure	Computer Scientist Computer Engineer System Analyst
Programming	Computer Programmer
Computer Operations	Computer Operator Data Encoder Data Entry Operator Computer Librarian
Others	Database Administrator

	Network or Systems Administrator Computer Support Specialist Telecommunication Specialist Computer Security Specialist Web Developer/ Web Master Application Specialist Application Architect Client/Server Quality assurance Analyst Client/Server Tester Hardware/ Software Test Engineer Quality Assurance Analyst Software Certification Specialist
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ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

1. W

LESSON 5: EVOLUTION OF COMPUTING

OVERVIEW:

This lesson discusses the development of computers.

LEARNING OUTCOME:

At the end of the lesson, the student can state the development of computers.

MATERIALS NEEDED:

Computer, Projector, Video presentation

DURATION:

240 minutes

LESSON PROPER:

HISTORY OF COMPUTERS

2000 BC	Abacus: The abacus, the first known calculator, was invented in Babylonia
1801	Joseph-Marie Jacquard: Joseph-Marie Jacquard invents an automatic loom controlled by punched cards
1822	Charles Babbage: Charles Babbage designs his first mechanical computer
1834	Analytical Engine: The Analytical Engine was invented by Charles Babbage
1890	Herman Hollerith: Herman Hollerith invents a counting machine which increment mechanical counters
1937	Alan Turing: Alan Turing develops the concept of a theoretical computing machine
1943	- Enigma: Adolf Hitler uses the Enigma encryption machine - Colossus: Alan Turing develops the the code-breaking machine Colossus
1945	ENIAC: John Presper Eckert & John W. Mauchly: John Presper Eckert & John W. Mauchly develop the ENIAC (Electronic Numerical Integrator and Computer)
1951	UNIVAC: UNIVAC I (UNIVERSal Automatic Computer I) was introduced - the first commercial computer made in the United States and designed principally by John Presper Eckert & John W. Mauchly

	EDVAC: The EDVAC (Electronic Discrete Variable Automatic Computer) begins performing basic tasks. Unlike the ENIAC, it was binary rather than decimal
1955	Bell Labs introduces its first transistor computer.
1958	Silicon chip: The first integrated circuit, or silicon chip, is produced by the US Jack Kilby & Robert Noyce
1963	The Computer Mouse: Douglas Engelbart invents and patents the first computer mouse (nicknamed the mouse because the tail came out the end)
1973	Personal computer: The minicomputer Xerox Alto (1973) was a landmark step in the development of personal computers
1975	- Portable computers: Altair produces the first portable computer - Microsoft Corporation: The Microsoft Corporation was founded April 4, 1975 by Bill Gates and Paul Allen to develop and sell BASIC interpreters for the Altair 8800
1976	Apple: Apple Computers was founded Steve Wozniak and Steve Jobs
1977	Apple Computer's Apple II, the first personal computer with color graphics, is demonstrated

ACTIVITY:

Video Viewing

EVALUATION:

W

LESSON 6: Key Components of a Computer System

OVERVIEW:

This lesson discusses the definition of a computer.

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. Be aware of the concepts of computing
2. Identify types of computer
3. Classify computers according to criteria
4. State capabilities and limitations of computer
5. Differentiate the generations of computer

MATERIALS NEEDED:

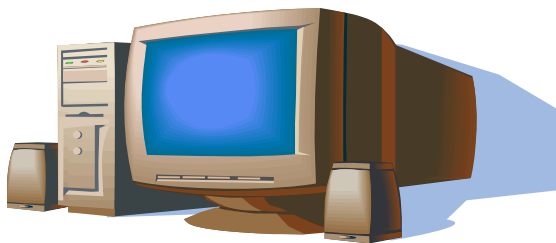
Computer, Projector, Presentation Slides

DURATION:

240 minutes

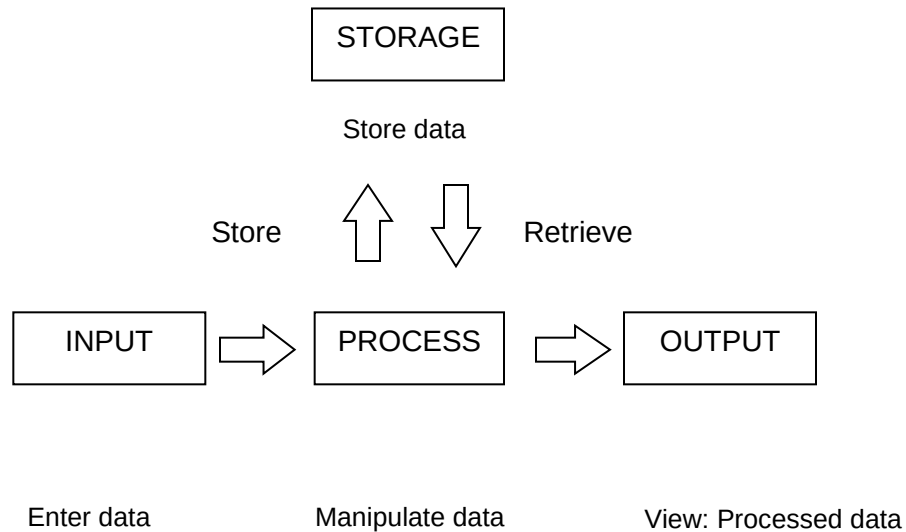
LESSON PROPER:

WHAT IS COMPUTER?



- It is an electronic device which consists of several components that together provide the capability of executing stored programs.
- Is a [programmable machine](#) designed to sequentially and automatically carry out a sequence of arithmetic or logical operations
- It is an electronic, programmable device that can store, retrieve and process data.

FUNCTIONS OF COMPUTER



ELEMENTS OF COMPUTER

- **Hardware.** Hardware are the physical equipment that you can feel and touch.
- **Software.** Software are programs or instructions that a computer must follow to do its task.
- **People.** They use and operate the computer for a variety of applications.
- **Data.** Data include texts and numbers, sounds, images and video that you input into the computer for processing.
- **Connectivity.** Connectivity means that a computer must be connected to other computers to share and access data.
- **Procedure.** Procedures are the step by step instructions to follow when using the hardware, software and data.

Advantages of Computer

Computers are becoming more and more popular tool of people in their homes, schools, offices and almost everywhere. They have the following capabilities:

1. **Speed.** Computer operates on data and commands at incredibly fast speed. In one second, billions or trillions of operations are done by the computer. The speed and capabilities of computers are getting faster and better every time a new model is made.
2. **Storage.** Computer can store enormous amount of data and information in its memory. These information are easily transferred and used anytime and anywhere. Information are easily and cheaply shared and stored. Storage

capacities of computers are getting larger and cheaper, in most cases they become smaller in size as technology advances.

3. **Reliability.** Computers made of modern technology rarely breakdown and when they do, they are easily repaired. They could operate the whole day, even months and years without interruption in some applications. New methods and components have no more moving parts, and sturdy, making them less prone to damage.
4. **Consistent.** If you input the same data into the computer using the same program, it will give you the same result all the time. If you input incorrect data and program, it will also give you incorrect results, this being known as computer phrase “garbage in, garbage out”. Computers are used in mass production of goods since they are able to produce the same results and specifications.
5. **Communicate.** People can connect with others by the use of computers. Computers have made it possible to talk and see other people over the internet. Information in all forms are shared faster and easier as well. Internet connection allows access to information and has made every home a virtual library.

Disadvantages of Using the Computer

The use of computer also presents some dangers for people and the environment.

1. **Safety.** You should be careful in sharing personal information on the internet and avoid instances wherein you will be indentified and your residence known. There are people with evil intention and have caused harm to a lot people already, using information taken from the internet, examples are stolen and fraudulent credit cards.
2. **Health Risks.** Long and improper use of the computer can lead to injuries and hazards to the user. The pain or injury may affect the eyes, hands, wrists, necks and back. Some injuries require surgery and some may become permanent. Proper posture and use of properly designed tables, chairs and equipment help to minimize these risks.
3. **Environment.** Old computers must be disposed off properly. Computer parts could release toxic or dangerous materials into the environment. Heat generated by the computer maybe small but if there are so many they could add to pollution and global warming, contributing to an already fragile worldwide environmental condition. As an example, the liquid crystals inside LCD monitors are poisonous.
4. **Use of power.** Computers need electricity to run. Computers only need a small amount of electric power but with many computer around, this would result to high total electricity consumption. Power is an expense, also a contributor to environmental damage.
5. **Privacy Violation.** Personal information when stored in a computer can be viewed and stolen. Criminal have used other people’s information to commit crimes.

Capabilities of Computer System

1. They can do repetitive and routine work.
2. They have the speed to process voluminous data in a flash.
3. Computers are reliable and accurate.
4. They can store and recall tremendous amount of information.
5. Computers have self-checking capability.
6. Computers can be self-operating.
7. They can do remote processing.

Limitations of Computer System

1. They are always dependent on instructions and data given through the program made by the programmer.
2. They can never generate information on their own.
3. They can never correct wrong instructions.
4. They cannot decide if they were not programmed for certain specific tasks.
5. They are vulnerable to virus attack.

CLASSIFICATIONS OF COMPUTER

1. **According to Capacity** with their ability to perform a task, as well as maximum number of programs they can run
 - a. Microcomputer
 - I. Desktop Computer: a personal or micro-mini computer sufficient to fit on a desk.



- II. Laptop Computer: a portable computer complete with an integrated screen and keyboard. It is generally smaller in size than a desktop computer and larger than a notebook computer.



- III. Palmtop Computer/Digital Diary /Notebook /PDAs: a hand-sized computer. Palmtops have no keyboard but the screen serves both as an input and output device.



b. Minicomputer

A midsized computer. In size and power, minicomputers lie between *workstations* and *mainframes*. In the past decade, the distinction between large minicomputers and small mainframes has blurred, however, as has the distinction between small minicomputers and workstations. But in general, a minicomputer is a multiprocessing system capable of supporting from 4 to about 200 users simultaneously.



c. Mainframe

A very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously. In the hierarchy that starts with a simple microprocessor (in watches, for example) at the bottom and moves to supercomputers at the top, mainframes are just below supercomputers. In some ways, mainframes are more powerful than supercomputers because they support more simultaneous programs. But supercomputers can execute a single program faster than a mainframe.



d. Super computer

The fastest type of computer. Supercomputers are very expensive and are employed for specialized applications that require immense amounts of mathematical calculations. For example, weather forecasting requires a supercomputer. Other uses of supercomputers include animated graphics, fluid dynamic calculations, nuclear energy research, and petroleum exploration.

The chief difference between a supercomputer and a mainframe is that a supercomputer channels all its power into executing a few programs as fast as possible, whereas a mainframe uses its power to execute many programs concurrently.



2. **According to Purpose** their goals and aims for a variety of tasks
 - a. General Purpose Computers
 - b. Special Purpose Computers
3. **According to Type of Processing** which are their special uses in different fields of work
 - a. Analog

An analog computer is a form of computer that uses *continuous* physical phenomena such as electrical, mechanical, or hydraulic quantities to model the problem being solved.



Speedometer

b. Digital

A computer that performs calculations and logical operations with quantities represented as digits, usually in the binary number system.



c. Hybrid

A combination of computers those are capable of inputting and outputting in both digital and analog signals. A hybrid computer system setup offers a cost effective method of performing complex simulations.



4. **According to Generation** – main components of the computer

- a. 0th Generation – mechanical computer
- b. 1st Generation – vacuum tubes
- c. 2nd Generation – Transistors
- d. 3rd Generation – Small Scale to Medium Scale Integrated Circuits
- e. 4th Generation – Large Scale to Very Large Scale Integrated Circuits

ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

LESSON 7: Key Components of Operating System

OVERVIEW:

This lesson covers explanation about Operating System.

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. Define operating system
2. Explain the significance of operating system
3. Identify different Operating systems
4. Familiarize themselves with the OS environment

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

Operating System

Operating system (OS) is software that acts as an interface between users of a computer system and the computer hardware.

Ex. Windows 7, Mac OS, Linux, Ubuntu, MS DOS, Jelly Bean,

Functions of the Operating System

1. Controls and coordinate the use of the hardware among various application programs for various reasons.
2. Provides the means for proper use of resources in the operation of the computer system. It provides an environment within each other programs can do useful work.
3. Acts as the manager of the resources of the computer system (hardware and software) and allocated them to specific programs and users as necessary for their tasks. Since there may be many, possibly conflicting, requests for

resources, the OS must decide which requests are allocated resources to operate the computer system fairly and efficiently.

4. **Control program** that control the execution of user programs to prevent errors and improper use of the computer. It is especially concerned with the operation and control of the I/O devices. For instance, it may terminate programs that are not responding.

Services of OS

1. **Program Execution** – the manner by which program are run in the machines and monitoring the way they perform as coded.
2. **Input/Output Operations** – processing signals coming from input devices to create a corresponding signal for the use of output devices.
3. **File System Manipulation/File Management** – handling of files and files system: *creating, copying, moving, renaming, and deleting* files and folders.
4. **Error Detection and Protection** - Monitoring errors in systems and applications programs and hardware, preventing them use various utilities.
5. **Resource Allocation and Accounting** – Allocating memory for various programs and optimizing hard disk space for better use.

Common Graphical User Interface

Icon – is the graphic symbol/ image that represent objects like files, folders, programs, devices and shortcuts

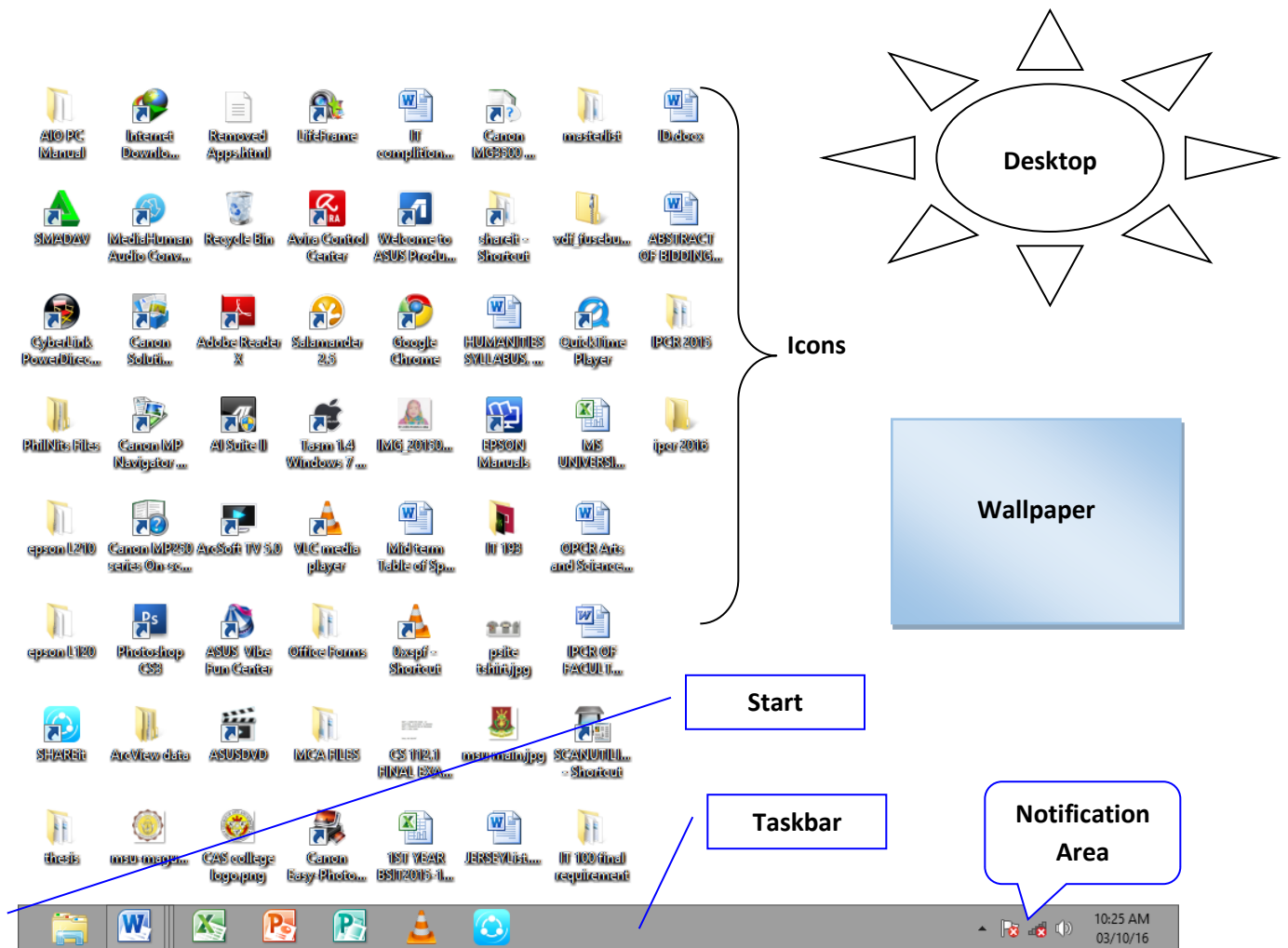
Desktop – is the main interface of the computer system. It is the entire area that we see on screen. It is also called the work area.

The backdrop on the screen is commonly called the wallpaper.

Start – is the button to press to begin work in windows. It allows you to open programs and data.

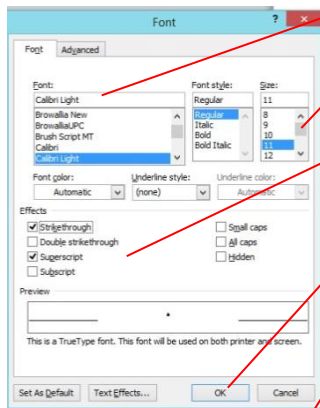
Taskbar – shows all programs that are currently open.

Notification Area – composed of different icons that display the status of a program and of the computer.



Windows Objects

Dialog Boxes – are main interface between users and programs.



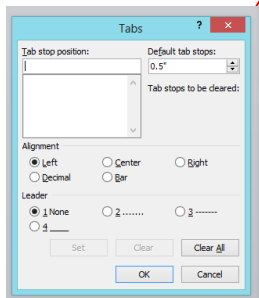
Textbox – along box where textual data are entered

Scroll bars – (vertical and horizontal) are used to view/ display the remaining contents of windows.

Check box – enables or disables certain capabilities. Can also be used to provide choices. Multiple choices can be selected.

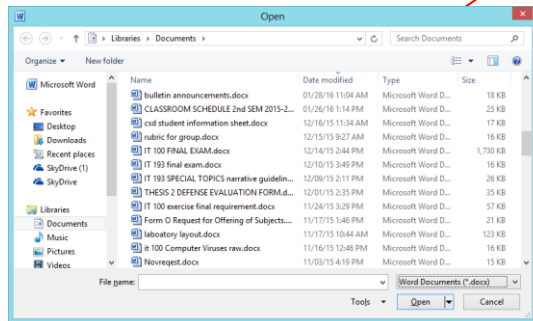
Command buttons – represent commands that are activated when clicked. (OK, Cancel)

Option buttons – group of tiny labelled circles that provides choices. Single choice can be selected.



List boxes – a space that displays a list of related objects.

Drop-down list boxes – appear like text box but contain a drop-arrow button at the right.



ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

W

LESSON 8: Malware

OVERVIEW:

This lesson defines the different malicious software that attacks our software and files.

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. Identify different malwares
2. Recognize malware behavior
3. protect computers from malwares from spreading and infections.

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

Computer Security

Cyberthreats

1. **Denial of Service Attacks (DoS) or Distributed Denial of Service Attack (DDoS)** - form of technological assault that consists of making repeated fraudulent requests of a computer system or network, thereby overloading it and denying legitimate users access to it.
2. **Malware** (Malicious Software)
 - **Virus** – attaches itself to a program or file enabling it to spread from one computer to another, leaving infections as it travels. It can cause unexpected and often undesirable effects such as destroying or corrupting data, or shut down entire computer system.
 - **Worm** – a program the repeatedly copies itself into a computer's memory or onto a disk until no space is left. It can shut down computers and networks.
 - **Trojan Horse** – a program that pretends to be a useful program, such as games that carry viruses or destructive instructions. It can cause trouble without your knowledge such as allow installation of backdoor programs,

illegal programs that allow illegitimate users to take control of your computer without your knowledge.

3. **Rootkits** – a program intended for operating system. This gives an attacker “super power” over computers – for example, the ability to steal sensitive personal information.

Root – an account for system administration

Kit – a malware introduced into the computer

4. **Zombies** and **Bots** – hijacked computers.

- Botmaster uses malware to hijack hundreds to many thousands of computers and is able to remotely control them all, including the ability to update malware and to introduce other programs such as spyware.

Botnet is a network of zombies or bots.

5. **Ransomware** – also known as cryptovirus. Ransomware holds the data on a computer or the use of the computer hostage until a payment is made.

6. **Spyware** – is a surveillance tool that spies on computer users and steals their information.

- Keylogger – secretly harvest every keystroke that a computer user makes and so steals sensitive data for profit.

7. **Time, Logic, and Email Bombs**

- **Time bomb** – a malware programmed to go off at a particular time or date.
- **Logic bomb** – is detonated when a specific event occurs
- **Email bomb** – overwhelmed a person’s email account by surreptitiously subscribing to it by dozens of even hundreds of mailing lists.

Spread of Malware

- By Infected disks, flash drives, and USB thumb drives
- By Opening unknown email attachments
- By Clicking on infiltrated websites
- Through infiltrated Wi-Fi hotspots

Ways to Minimize Virus Attacks

- Don’t open, download, execute any files, email messages, or email attachments if the source is unknown or if the subject line of an email is questionable or unexpected.
- Delete all spam and email messages from strangers. Don’t open, forward or reply to such messages
- Use webmail sparingly
- Don’t start your computer with a flash drive, USB thumb drive or CD/DVD in place
- Backup your data files regularly and keep at least one backup device in a location separate from your computer

- Make sure you have virus protection software
- Update antivirus
- Scan CDs and drives before using them
- Report any suspected virus to antivirus company

ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

W

LESSON 9: Complex Security

OVERVIEW:

This lesson identifies different entity involve in the security of computer information and the ways to secure information.

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. Distinguish different types of hackers
2. Discuss measures to secure information

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

Cybervillains: Hackers and Crackers

“All crackers are hackers, but not all hackers are crackers”

Hackers – are computer enthusiasts, people who enjoy learning programming languages and computer systems who gain unauthorized access to computers or networks often just for the challenge of it.

Types of Hackers

- **Thrill-seeker hackers** – are hackers who illegally access computer systems for simply for the challenge of it.
- **White-hat hackers** – are usually computer professionals who break into computer systems and networks with the knowledge of their owners to expose security flaws that can be fixed.

Crackers – are malicious hackers who break into computers for malicious purposes – to obtain information for financial gain, shutdown hardware, pirate software, steal people’s credit information, or alter or destroy data.

Classes of Hackers

- **Script kiddies** – people who use downloadable software or source code to perform malicious break-ins

- **Hactivist** – “hacker activist” are people who break into computer system for politically or socially motivated purpose.
- **Black-hat hackers** – are those who break into computer systems to steal or destroy information or to use it for illegal profit.
- **Cyberterrorist** – are politically motivated persons who attack computer systems so as to bring physical or financial harm to a lot of people or destroy a lot of information

Safety

1. **Antivirus software** – scans a computer’s hard disk, other designated storage devices and main memory to detect viruses and sometimes to destroy them.
2. **Firewall** – is a system of hardware or software that protects a computer or a network from intruders.
3. **Passwords** –
4. **Biometric authentication** –
 Biometric – the science of measuring individual body characteristics
 - i. Hand-geometry systems/ full-hand palm scanners
 - ii. Fingerprint scanners
 - iii. Iris-recognition systems
 - iv. Face recognition systems
 - v. Voice recognition systems
5. **Encryption** – is the process of altering readable data into unreadable form to prevent unauthorized access.

ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

W

LESSON 10: Network, Internet, and Internet Protocols

OVERVIEW:

This lesson talks about interconnection of the computers and ways of information interchange and mining.

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. Define basic computer network terminologies
2. Outline basic network configurations
3. Define internet
4. Learn how to connect to the internet
5. Identify the services provided by the internet
6. Learn how to search for information on the web

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

Basic Computer Networks

What Is a Network?

- A **network** is nothing more than two or more computers connected to each other so that they can exchange information, such as e-mail messages or documents, or share resources, such as disk storage or printers. In most cases, this connection is made via electrical cables that carry the information in the form of electrical signals.
- A **network** also requires special software to enable communications.

Network building blocks

Small or large, all networks are built from the following basic building blocks:

- **Client computers:** The computers that end users use to access the resources of the network. Client computers are typically located on users' desks. Client computers are sometimes referred to as workstations.

- **Server computers:** Computers that provide shared resources, such as disk storage and printers, as well as network services, such as e-mail and Internet access. Server computers typically run a specialized network operating system such as Windows Server 2008 or 2003, NetWare, or Linux, along with special software to provide network services. For example, a server may run Microsoft Exchange to provide e-mail services for the network, or it may run Apache Web Server so that the computer can serve Web pages.

- **Network Interface Card (NIC):** An interface that's installed in a computer that enables the computer to communicate over a network. Almost all NICs implement a networking standard called Ethernet. Note that even though the "C" in "NIC" stands for "card," the NIC in most computers isn't actually a separate card. Instead, the NIC is an integral part of the computer's motherboard. Every client and every server computer must have a NIC in order to be a part of a network.

- **Cable:** Computers in a network are usually physically connected to each other using cable. Although several types of cable have been popular over the years, the most commonly used cable today is called **twisted pair**, also known by its official designation **10BaseT**. Another type of cable commonly used is **coaxial**, also called **10Base2**. For high-speed network connections, **fiber-optic** cable is sometimes used.

- **Switches:** Network cable usually doesn't connect computers directly to each other. Instead, each computer is connected by cable to a device known as a switch. The switch, in turn, connects to the rest of the network. Each switch contains a certain number of ports, typically 8 or 16. Thus, you can use an eight-port switch to connect up to eight computers. Switches can be connected to each other to build larger networks.

(Older networks may use a more primitive type of device called a hub instead of a switch. A hub provides the same function as a switch, but it isn't as efficient.

- **Wireless networks:** In many networks, cables and switches are making way for wireless network connections, which enable computers to communicate via radio signals. In a wireless network, radio transmitters and receivers take the place of cables. The main advantage of wireless networking is its flexibility. With a wireless network, you don't have to run cables through walls or ceilings, and your client computers can be located anywhere within range of the network broadcast.

- **Network software:** Although network hardware is essential, what really makes a network work is software. A whole bunch of software has to be set up just right in order

to get a network working. Server computers typically use a special network operating system (also known as a NOS) in order to function efficiently, and client computers need to have their network settings configured properly in order to access the network.

Networks are about sharing three things: information, resources, and applications.

1. **Sharing information:** Networks allow users to share information in several different ways. The most common way of sharing information is to share individual files. For example, two or more people can work together on a single spreadsheet file or word-processing document. In most networks, a large hard drive on a central server computer is set up as a common storage area where users can store files to be shared with other users.

2. **Sharing resources:** Certain computer resources, such as printers or hard drives, can be set up so that network users can share them. Sharing these resources can result in significant cost savings. For example, it is cheaper to buy a single high-speed printer with advanced features such as collating, stapling, and duplex printing that can be shared by an entire workgroup than it is to buy separate printers for each user in the group. Hard drives can also be shared resources. In fact, providing users with access to a shared hard drive is the most common method of sharing files on a network. A computer whose main purpose in life is to host shared hard drives is called a file server.

3. **Sharing applications:** One of the most common reasons for networking in many businesses is so that several users can work together on a single business application. For example, an accounting department may have accounting software that can be used from several computers at the same time.

A **server**: is the network computer that contains the hard drives, printers, and other resources that are shared with other network computers.

a. Any computer that's not a server is called a client.

b. The most powerful and expensive computers in a network are the servers. This fact makes sense because every user on the network shares the server's resources.

c. The cheaper and less powerful computers in a network are the clients. Clients are the computers used by individual users for everyday work. Because clients' resources don't have to be shared, they don't have to be as fancy.

d. Most networks have more clients than servers. For example, a network with ten clients can probably get by with one server.

e. In some networks, a clean line of segregation exists between servers and clients. In other words, a computer is either a server or a client, and not both. A server can't become a client, nor can a client become a server.

Types of Networks

- **Local Area Networks(LAN):** A local area network, or LAN, is a network in which computers are relatively close together, such as within the same office or building. A LAN can, in fact, contain hundreds of computers.
- **Wide Area Networks(WAN):** A wide area network, or WAN, is a network that spans a large geographic territory, such as an entire city, region, or even an entire country. WANs are typically used to connect two or more LANs that are relatively far apart. For example, a WAN may connect an office in San Francisco with an office in New York.
- **Metropolitan Area Networks(MAN):** A metropolitan area network, or MAN, is a network that's smaller than a typical WAN but larger than a LAN. Typically, a MAN connects two or more LANs within a same city but are far enough apart that the networks can't be connected using a simple cable or wireless connection.

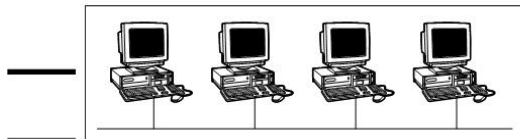
Network Topology

The term network topology refers to the *shape* of how the computers and other network components are connected to each other. There are several different types of network topologies, each with advantages and disadvantages.

- **Node:** A node is a computer that is connected to the network. Network topology deals with how the nodes of a network are connected to each other.
- **Packet:** A packet is a message that is sent over the network from one node to another node. The packet includes the address of the node that sent the packet, the address of the node the packet is being sent to, and data.

Kinds of Network Topology:

1. **Bus topology** - nodes are strung together in a line, as shown. Bus topology is commonly used for LANs.

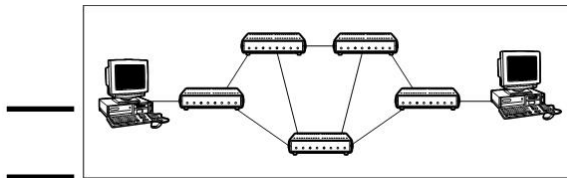


In a bus topology, every node on the network can see every packet that's sent on the cable. Each node looks at each packet to determine whether the packet is intended for

it. If so, the node claims the packet. If not, the node ignores the packet. This way, each computer can respond to data sent to it and ignore data sent to other computers on the network.

If the cable in a bus network breaks, the network is effectively divided into two networks. Nodes on either side of the break can continue to communicate with each other, but data can't span the gap between the networks, so nodes on opposite sides of the break can't communicate with each other.

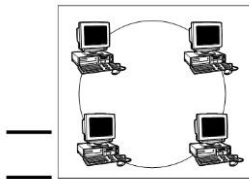
2. **Star topology** - each network node is connected to a central device called a hub or a switch. Star topologies are also commonly used with LANs.



If a cable in a star network breaks, only the node connected to that cable is isolated from the network. The other nodes can continue to operate without interruption — unless, of course, the node that's isolated

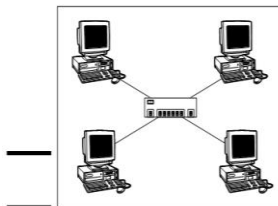
because of the break happens to be the file server.

3. **Ring topology** - packets are sent around the circle from computer to computer.



Each computer looks at each packet to decide whether the packet was intended for it. If not, the packet is passed on to the next computer in the ring.

4. **Mesh topology** - has multiple connections between each of the nodes on the network. The advantage of a mesh topology is that if one cable breaks, the network can use an alternative route to deliver its packets.



INTERNET

Internet – world's largest computer network

www/world wide web/w3 – the link which established the physical connection of documents from one computer to another that formed like a spider web.

Internet Requirements

1. Hardware – computer
2. Phone Line Connection
3. Internet Service Provider (ISP) – is the company that sells access to the internet
4. Web browser – a graphical interface software

Web servers – the computer residence of a collection of information or files publicly available

Website – the collection of linked web pages with a common topic or focus

Uniform Resource Locator (URL) – website address

Parts:

1. transfer protocol specifications – http, ftp, ...
2. domain - .com, .edu., .org, .net, .gov, ...
3. document name

example:

http://www.msumaguindanao.edu.ph/main.html

transfer protocol specification domain name document name

Internet Services

1. e-mail (electronic mail) – sending messages is done from one computer to another parts:

1. account name – name used to log on to your internet account
- username, user id, etc...
2. @ sign – used as a pointer to separate username from the domain name
3. domain name – provides the address of the ISP provider or online service

ex:

msumaguindanao@yahoo.com

account name @ sign domain name

2. Chat and Talk

Chat – is a system that lets you communicate with others in a public arena.

Talk – “instant messaging”

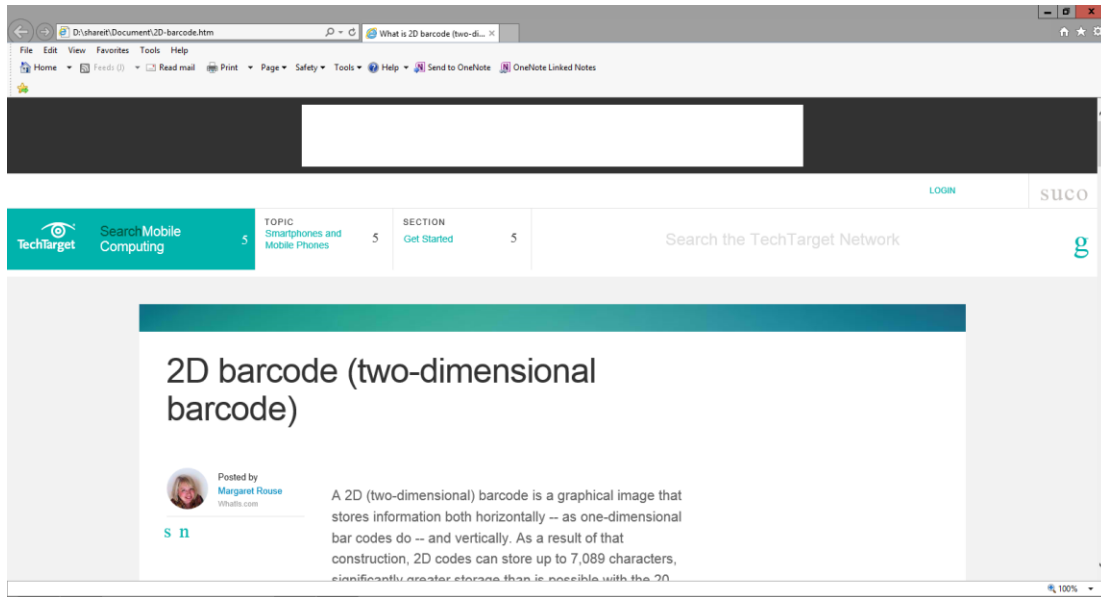
- allows two people to chat privately online.

3. Internet Phones

4. Newsgroup - like a giant electronic bulletin where messages are posted for everybody to read and respond to.

5. Information Retrieval - access to thousands of online library information, databases open to the public and high quality software made available as freeware, shareware or open source.

Browser Environment



Common Parts of Web Browser Screen

Title bar – displays the current program and web page title

Menu bar – displays the commands like File, Edit, ...

Toolbar – provides shortcuts to the standard buttons toolbar

Address bar – displays current address

Links bar – provides shortcut link to other web pages

Status bar – provides information on the status connection to the current address

Browser window – displays area or work area.

Standard buttons

Back – previous web page

Forward – moves the next previously visited page if back button was clicked.

Stop – interrupts the transfer of a new web page

Refresh – reloads the current web page

Home – displays the user defined start page

Search – opens a search engine for finding a specific topic on the web

Favorites – opens a list of web pages saved as favorites

History – opens a list of web pages previously visited

Print – prints the current web page

Searching/ Mining the Internet

Search Engine - searches for keywords and returns a list of documents where the keywords can be found

ex: Yahoo, Google, Excite, AltaVista

Tips for Effective Searching

- Beware of using slang or partial words.
you may have some luck, but your results may be too broad.
ex, flick instead of movie,
Macs instead of Macintoshes
- Correct Spelling, Pluralization and Capitalization.
ex, good will, goodwill, Goodwill, Goodwills, and Good Will
- Use nouns as keywords.
Words such as “a” and “the”, pronouns such as “she” and “he”, and prepositions such as “over” and “to” slow down the search engine and often produce irrelevant results.
- Enter as many keywords as possible.
1 to 2 words is not enough for an effective query as it will likely yield millions of results. Choose 5 to 7 keywords that narrow your search field.
- Use exact phrases when searching.
Phrases that must appear in the documents you trying to locate. Quotation marks (“”) should surround each phrase.
- Understand the Boolean Basics.
“Boolean searching” is technique that can narrow your search results and increase the likelihood that relevant results will be displayed.
Boolean commands, or operators (AND, OR, and NOT)
 - + or **AND** command will only return documents that contain all the keywords listed.
 - OR** command will return documents that contain any of the keywords.
 - or **NOT** command will return documents that do not contain the keyword listed.
- Wildcards.
The percentage symbol (%) can be used to replace only one character. The asterisk (*) can be used to replace multiple characters.
ex, post* could produce postwar, postgame, or postmarital.
- Use the advanced search features.
most search engines do have an advanced search feature which can help you narrow the results.
- Get Help.
Go to your search engine's Help page for more specific information on a particular engine.
- Try Something New.
There are many different search engines on the Web. If you can't find what you are looking for using your favorite engine, go somewhere else.
- Bookmark it.

Add your favorite search engines to your Favorites to increase ease of use.

ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

W

LESSON 11: HTML and CSS

OVERVIEW:

LEARNING OUTCOME:

At the end of the lesson, the student can:

1. become familiar with basics of webpage design.
2. become familiar with the basics of HTML/ CSS and Javascript.

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

Hypertext Markup Language (HTML):

On the Internet, a website is represented by a URL (uniform resource locator). As mentioned in the class, the World Wide Web (WWW), also abbreviated as the web, is a system of interlinked hypertext documents that may be viewed through a web browser. We mentioned in the class lectures that Internet Explorer and Mozilla Firefox are examples of web browsers. To understand how WWW works, it's important to first define what hypertext documents are. We use the following definition of hypertext:

“Hypertext most often refers to text on a computer that will lead the user to other, related information on demand. Hypertext allows a form of user interface, which overcomes some of the limitations of written text. Rather than remaining static like traditional text, hypertext makes possible a dynamic organization of information through links and connections (called hyperlinks). Hypertext can be designed to perform various tasks; for instance when a user "clicks" on it or "hovers" over it, a bubble with a word definition may appear, a web page on a related subject may load, a video clip may run, or an application may open.” [\[Wikipedia\]](#)

HTML is a hypertext *markup* language that is used to describe the structure of a webpage using tags. A web browser, an application that understands HTML tags and displays web pages accordingly, utilizes the HTTP protocol to receive HTML pages from a web server (also called HTTP server).

It is important to note that the display of an HTML page in a browser is not the same text that we see if we were to edit the HTML page in a text editor. This is because the HTML file itself contains many instructions/ tags that are not displayed in the browser. The function of these tags/ information is simply to enable a browser to display the HTML file. This should become clear as we go through the exercises.

HTML Resources:

The following are some tutorials on the topic of HTML. Tutorials on advanced topics such as JavaScript can also be found at the following sites:

- 1) <http://w3schools.com/html/default.asp>
- 2) <http://inpics.net/htmlcss.html>

Lab Exercise:

Part 1---Creating directory/ index.html and format.css files

A web server is an application that runs on TCP port number 80. It simply opens up a socket on the server's IP address' port 80 and waits for HTTP request. Examples of web server application software are the Apache Software Foundation's [Apache](#) and Microsoft's [IIS](#).

A web server (like Apache and IIS) has to be told of certain parameters (called configuration). This configuration is usually done by the webmaster managing the website. The configuration parameters include 1) What IP address/ domain name is the server tied to; 2) What port number should be used (this is 80 by default)?; 3) What is the home directory of the web server; 4) What is the name of the default HTML file in a directory hosted on the web server.

When a request for a site is sent (without any document specified), the web server application sends the default file in the home directory of the web server. As an example, if we type <http://www.niit.edu.pk> in the address bar of Internet Explorer or Mozilla Firefox, the default page in the default directory is served. Note that in the mentioned example, only the domain (www.niit.edu.pk) is mentioned, but not the specific web page that should be fetched from this domain; in such cases, the 'default' files as configured for the webserver will be fetched. Generally, the default file names

start with **index** such as index.html, index.php, etc. *[PHP is a web scripting language that is outside the scope of this lab]*

In this part, we will create a directory in which we will save today's lab work. We will also learn how to create and save *index.html* and *format.css* files.

1. On the desktop, create a folder named ICTLab6

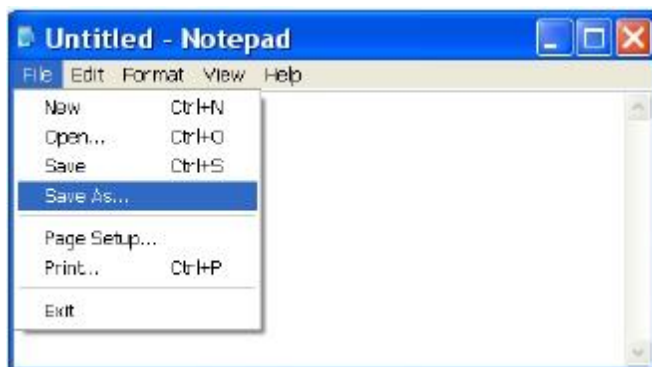
[Hint: this can be done by right clicking on the desktop, and then choosing **New** and then **Folder**]

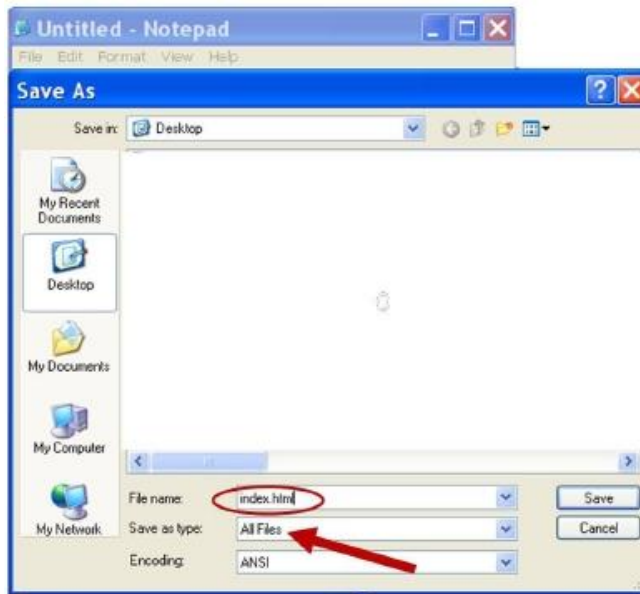
2. Create an index.html.

Open the Notepad application. [Start > Programs > Accessories > Notepad]

Save the empty file as index.html by clicking on the File menu of the Notepad.

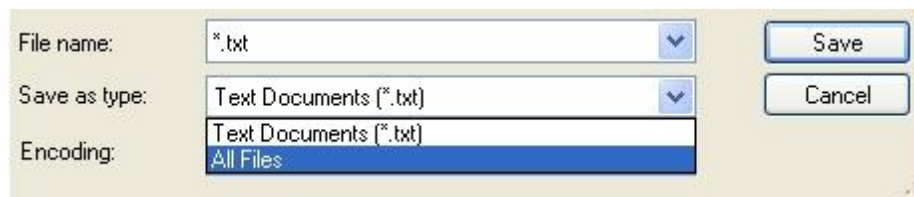
On the file menu, choose "Save As". The "Save As" window will appear as shown below:



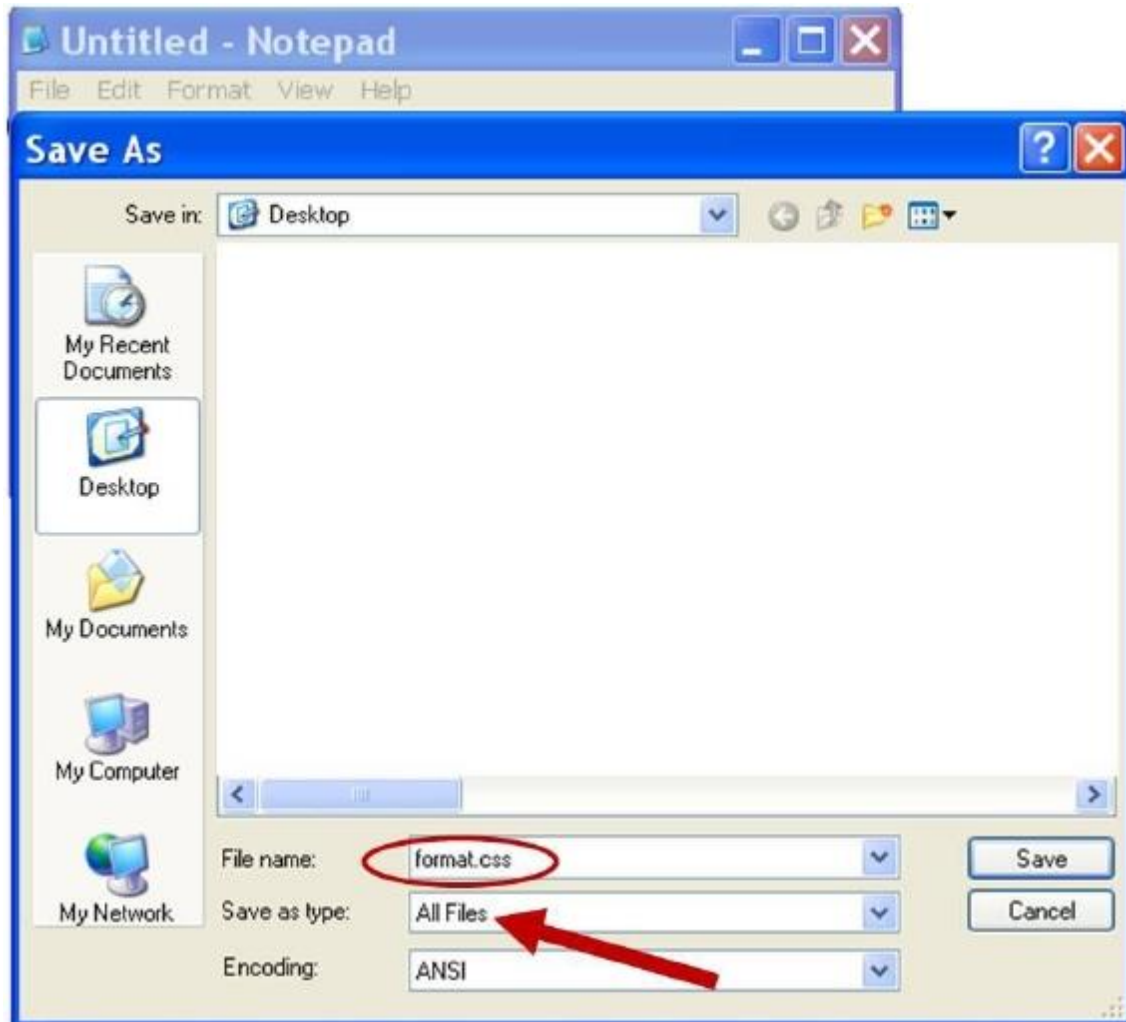


Save the empty file as index.html on the Desktop. [We will use the Desktop as HTML playground, and once the files have been finalized, then they will be transferred to the ICTLab6 folder.] To save the file as index.html, index.html must be written as the file name with the “Save as type” set to *All Files*.

Note: As explained the “Save as type” should be set to *All Files*, if the “Save as type” field is set to *Text Documents (*.txt)* then the file would be saved as index.html.txt and not as index.html.



3. In a similar manner, create the format.css file. *[We will later see how this file is used]*

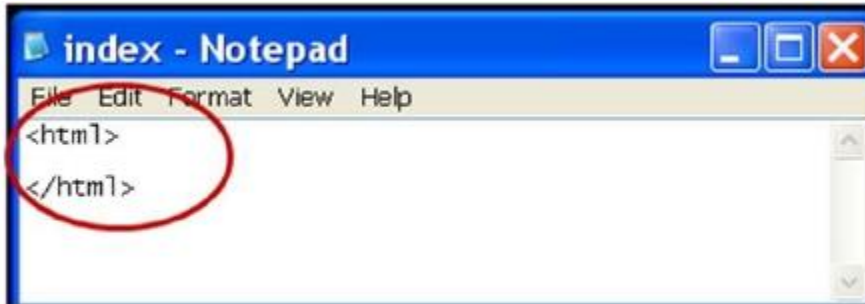


Part II---HTML basics

1. Writing the HTML basic tags

Open the index.html file on the Desktop, and enter the `<html>` and `</html>` tags on the file. Note that in HTML, tags often (but not always) follow the convention that there is an opening tag and a closing tag.

The opening tag in our context is `<html>` and the closing tag is `</html>`. These tags are used to inform the browser that the text between these tags is formatted according to HTML guidelines. The `<html>` tags are displayed in the figure below.



In the figure above, we do not have any HTML code in between the `<html>` tags. We will now start to code the HTML page.

2. Inserting the title of the web page

We will, first of all, learn the `<head>` tag and the `<title>` tag.

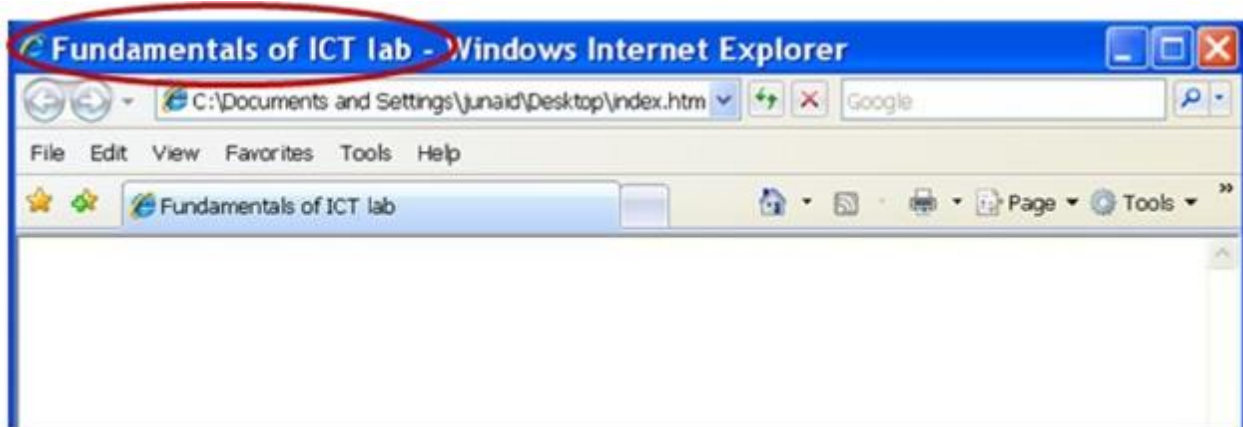
The **head** element (text between the `head` tags) contains general information, also called meta-information, about a document. The **title** (text between the `title` tags) is not part of the text of the document, but is a property of the whole document. It may not contain anchors, paragraph marks, or highlighting. The title may be used to identify the node in a history list, to label the window displaying the node, etc.

We will name the title of our webpage as Fundamentals of ICT lab. Note how the title text is enclosed in the **head** tags and the **title** tags.



Open the index.html file on the Desktop (modified as above) with Internet Explorer (this can be done by simply double clicking on the index.html file). You will find that the Internet Explorer displays "Fundamentals of ICT lab" as the title of the web page as configured in the figure above. The webpage is still empty as we have not added any text in the main page's body yet (we will do that soon).

For now, the index.html page should look like the following.



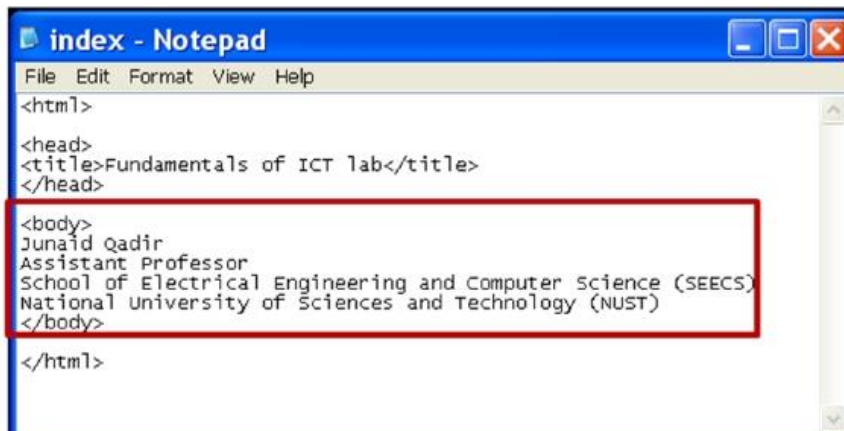
Copy the index.html into the ICT Lab0 folder on Desktop and rename it as **file.html**.

[Hint: a file can be renamed by selecting, right clicking, and choosing the rename option.]

3. Inserting text in a HTML page

We will now add text to the main body of the HTML page. This is done using the **<body>** and **</body>** tags. The text contained between the **<body>** and **</body>** tags is displayed by a browser unlike data that is contained between the **<head>****</head>** tags.

Add the text contained in the following figure in the index.html file on the Desktop. Save the file when done. The index.html file on Desktop should like the following.



After saving the index.html file on Desktop, open the file in a browser (by double clicking on the file). If the index.html file is properly configured, the browser display should be as follows. You will notice that all there is no line break, and all the text we had put between the **<body>** tags is on one line.

A browser has to be told by HTML tags where line breaks, new paragraphs, etc. should occur. Since we have not used those tags yet, all the text is on one line as shown in the figure below.

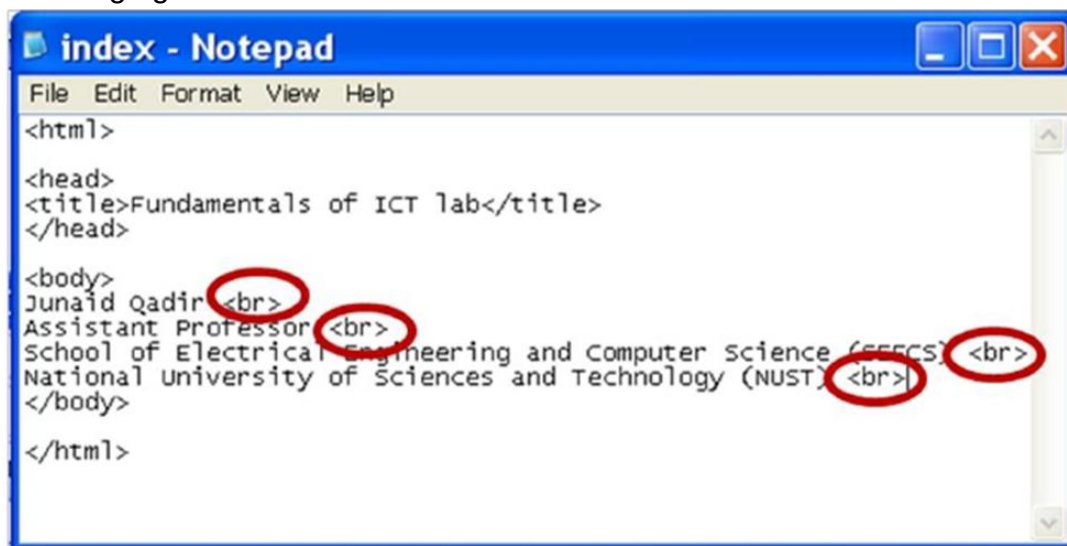


Save the index.html file on the Desktop in the ICTLab6 folder as **Allon1line.html**.

4. Inserting a line-break

We will now use the **
** HTML tag to introduce line breaks. This tag does not have a corresponding close tag and is used for introducing line breaks.

Open the index.html file on the Desktop and make the changes highlighted in the following figure.



After making the changes, save the index.html file and open it using a browser. The browser should display the page with proper line breaks as shown in the figure below. You will see that all the text is not on the same line any more.

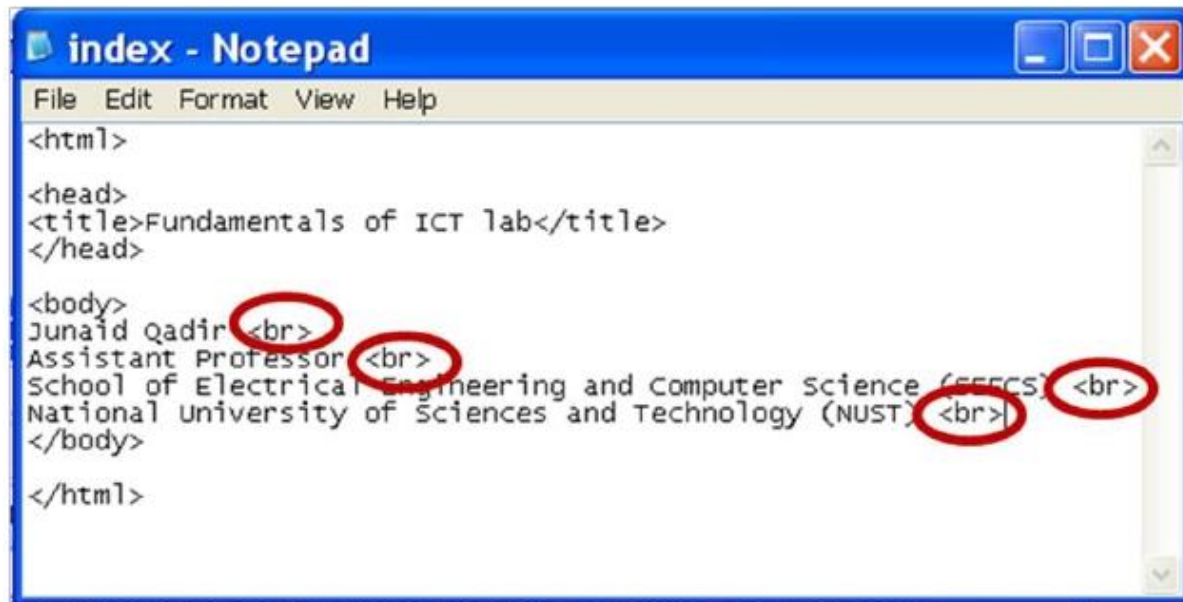


If you see the browser display as above, save the index.html file on the Desktop to the folder ICTLab6 as **Linebreak.html**.

5. Insert a paragraph

In the figure above, all the text is in a single paragraph (although separated by line breaks). To start a new paragraph, the HTML tag `<p>` should be used. The tag `<p>` is used to mark the start of a paragraph, and the end of paragraph is marked by `</p>`.

To observe how a paragraph is formatted in HTML, open the index.html file on the Desktop. Make the changes highlighted in the following figure and then save the file.



```
<html>
<head>
<title>Fundamentals of ICT lab</title>
</head>
<body>
Junaid Qadir <br>
Assistant Professor <br>
School of Electrical Engineering and Computer science (SEECs) <br>
National university of Sciences and Technology (NUST) <br>
</body>
</html>
```

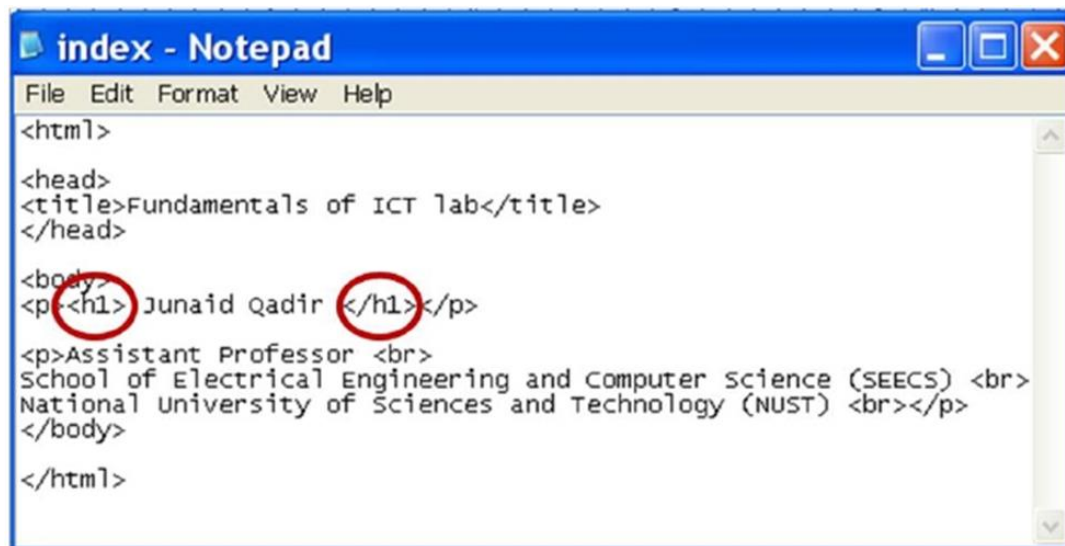
Open the index.html file on the Desktop (modified with the changes highlighted above) with a browser. You will see now that the web page is composed of two paragraphs. The second paragraph also has line breaks within the paragraph. The browser display should look like the following.



6. Inserting headings

In the index.html file on the Desktop, currently no text is highlighted as a heading. If certain text is to be a heading, the HTML tags `<h1></h1>`, `<h2></h2>`, `<h3></h3>`, `<h4></h4>` (and so on) may be used. The tag `<h1>` is used for bigger sized heading with the heading size decreasing for `<h2>` `<h3>` and so on.

Open the index.html file on the Desktop and make the changes highlighted in the following figure.



```
index - Notepad
File Edit Format View Help
<html>
<head>
<title>Fundamentals of ICT lab</title>
</head>
<body>
<p><h1> Junaid Qadir </h1></p>
<p>Assistant Professor <br>
School of Electrical Engineering and Computer science (SEECs) <br>
National University of Sciences and Technology (NUST) <br></p>
</body>
</html>
```

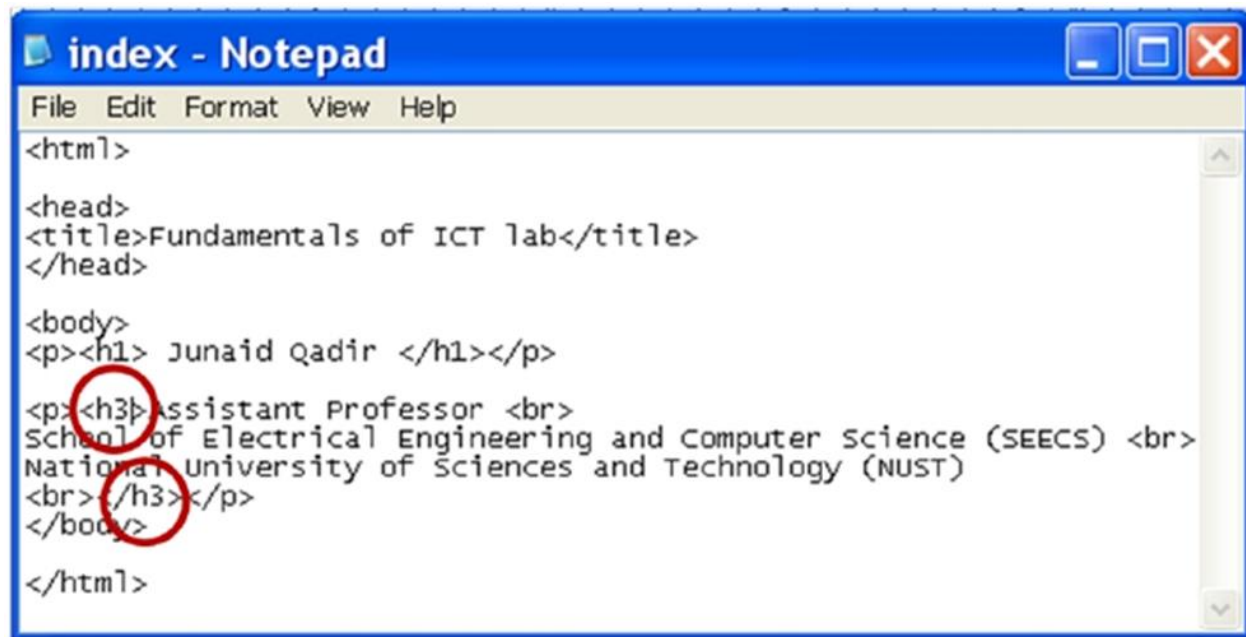
The first paragraph is highlighted using the `<h1>` tags.

After making the changes in the index.html file on the Desktop as shown above, save the file. Open the file now with a browser. The browser display should look like the following. You will note that the first paragraph now highlighted as a heading with a large sized font.



If the browser display is as shown above, save the index.html file (on the Desktop) in the ICT lab6 folder as **Headings1.html**

Open the index.html file on the Desktop and make the changes highlighted in the figure



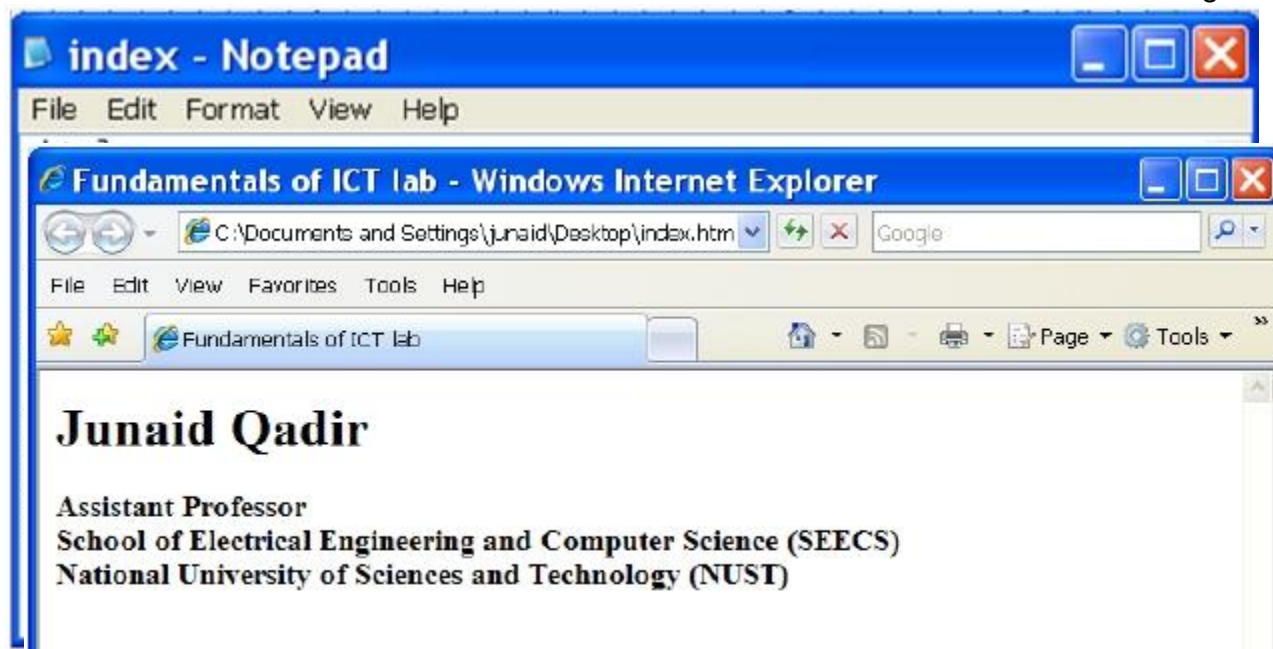
```
<html>

<head>
<title>Fundamentals of ICT lab</title>
</head>

<body>
<p><h1> Junaid Qadir </h1></p>
<p><h3>Assistant Professor <br>
School of Electrical Engineering and Computer Science (SEECS) <br>
National University of Sciences and Technology (NUST)
<br></h3></p>
</body>

</html>
```

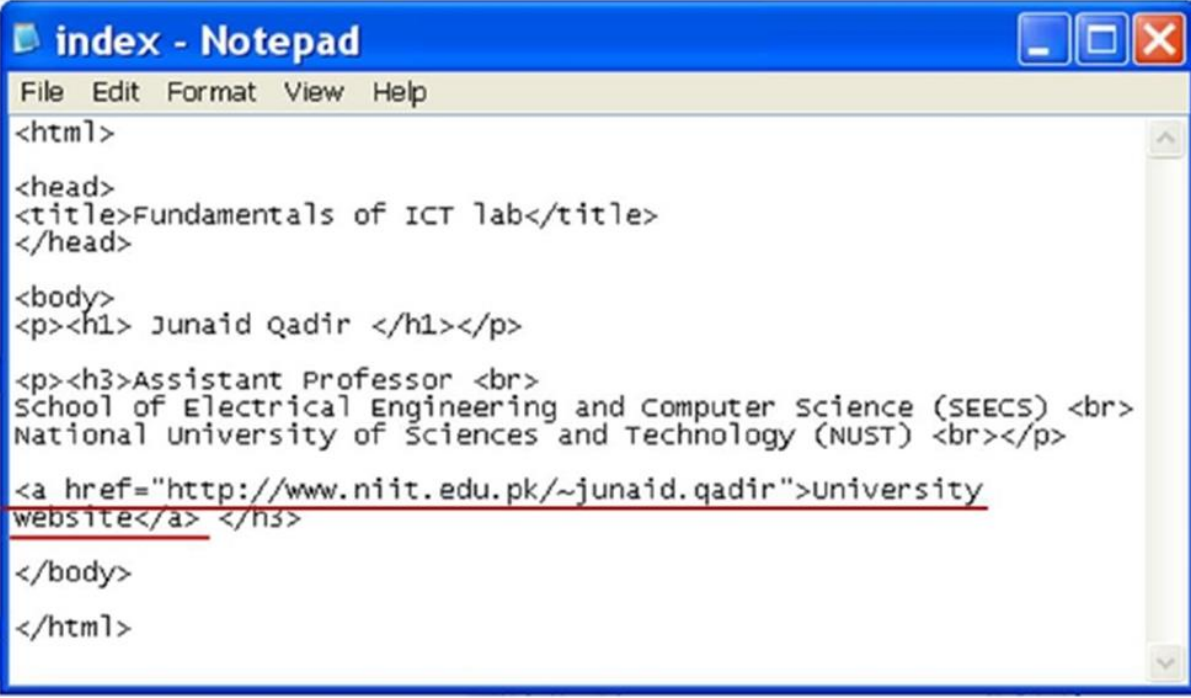
e



If the browser display is similar to as shown above, save the index.html file (on the Desktop) to the folder ICTLab6 as **Headings2.html**.

7. Inserting a hyperlink

We will now add a hyperlink on the index.html file. Open the index.html file on the Desktop and make the change highlighted in the following figure.



```
<html>
<head>
<title>Fundamentals of ICT lab</title>
</head>
<body>
<p><h1> Junaid Qadir </h1></p>
<p><h3>Assistant Professor <br>
School of Electrical Engineering and Computer Science (SEECs) <br>
National University of Sciences and Technology (NUST) <br></p>
<a href="http://www.niit.edu.pk/~junaid.qadir">university
website</a> </h3>
</body>
</html>
```

Save the file after making the change highlighted above. Open the file using a browser. You should see that the browser display is similar to as shown below. You will observe that a hyperlink to the specified URL is now added. Click on the URL and note where the target of the hyperlink.



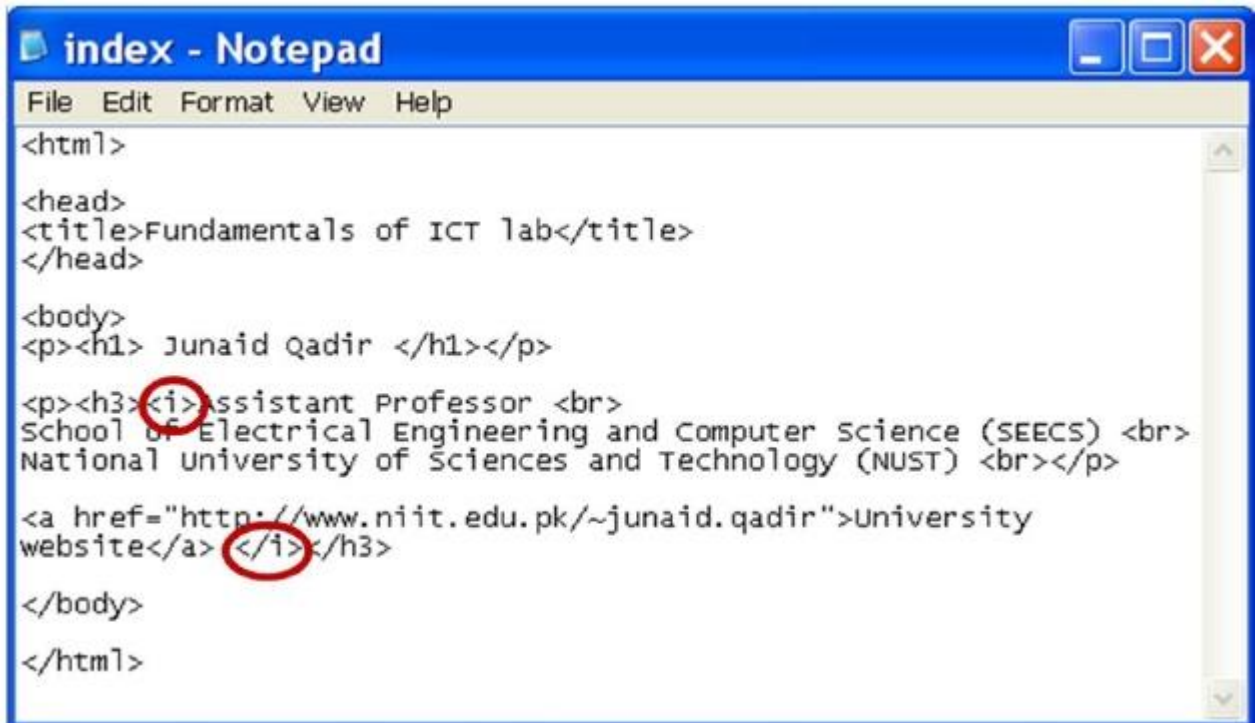
If the browser display is similar to as shown above, save the index.html file on the Desktop to the folder ICTLab6 as **Hyperlink.html**.

8. Making text italic, underlined, or bold

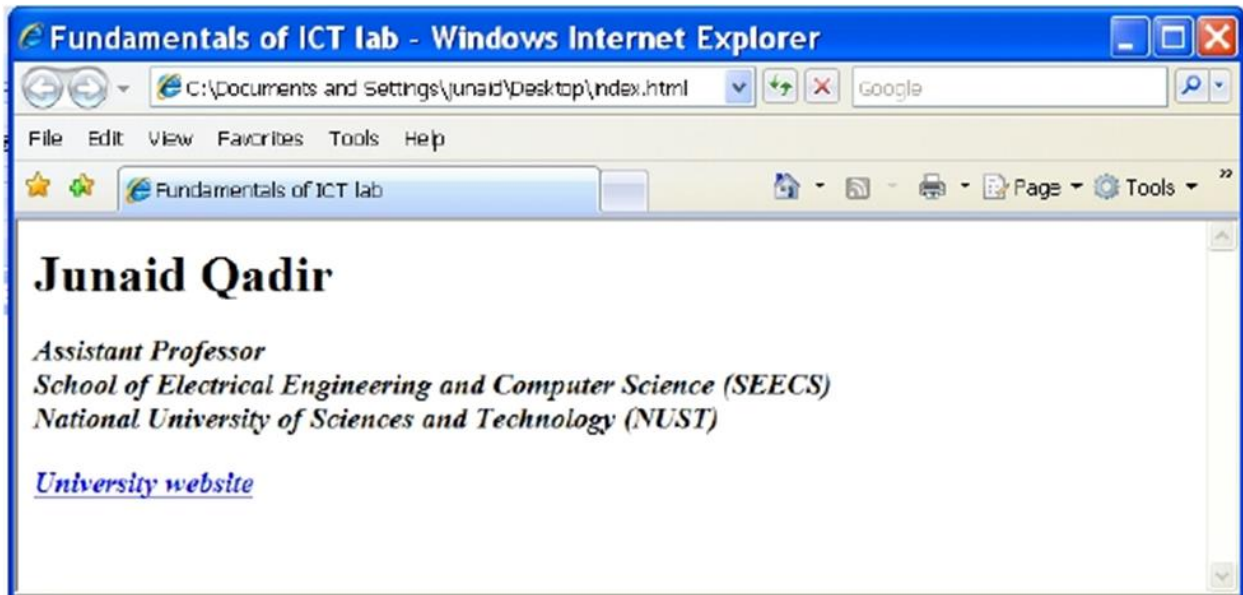
We will now see how a text can be *italicized*, underlined, or made **bold** in HTML. The tags for these tasks are, respectively, `<i></i>`, `<u></u>` and `<b></b>`.

Italicizing: To italicize a body of text, it must be enclosed in `<i> </i>` tags. To demonstrate this, open the index.html file on the Desktop and make the changes highlighted in the following figure.

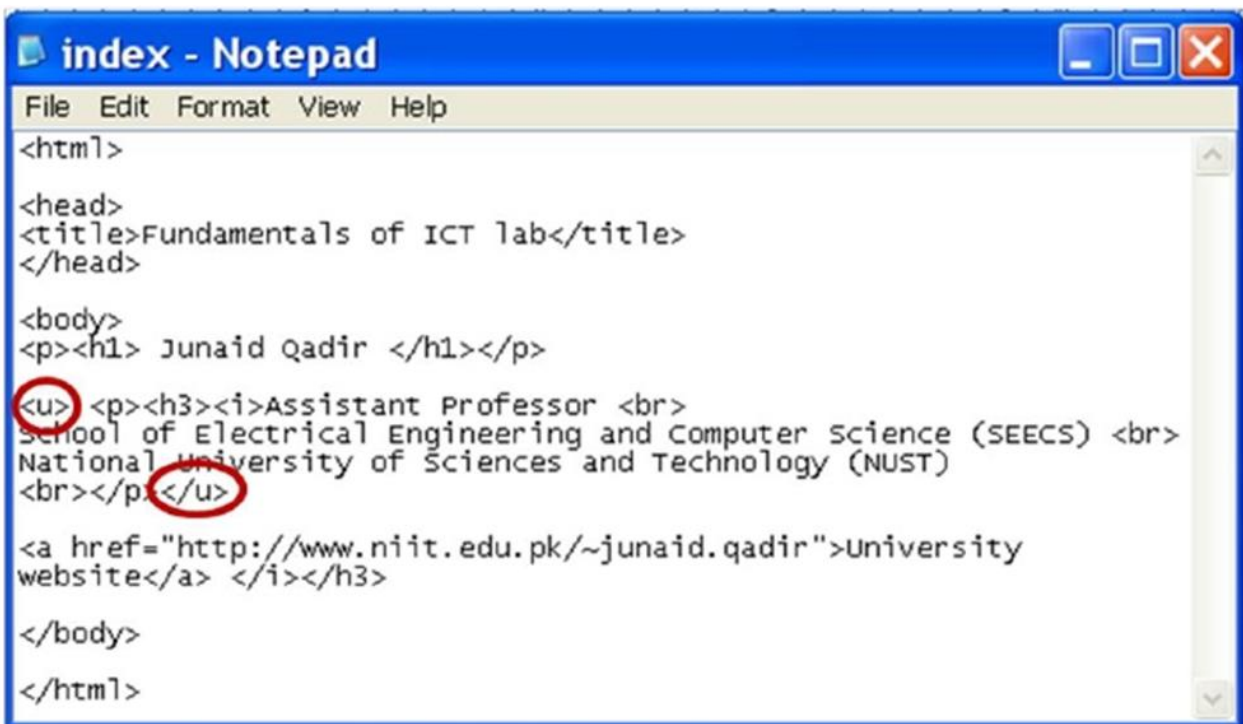
After making the changes shown above, save the file and then open it using a browser. You will observe that the text between the `<i>` and `</i>` tags is italicized. If the browser display is as shown below, save the index.html file in the ICTlab6 folder as **Italics.html**.



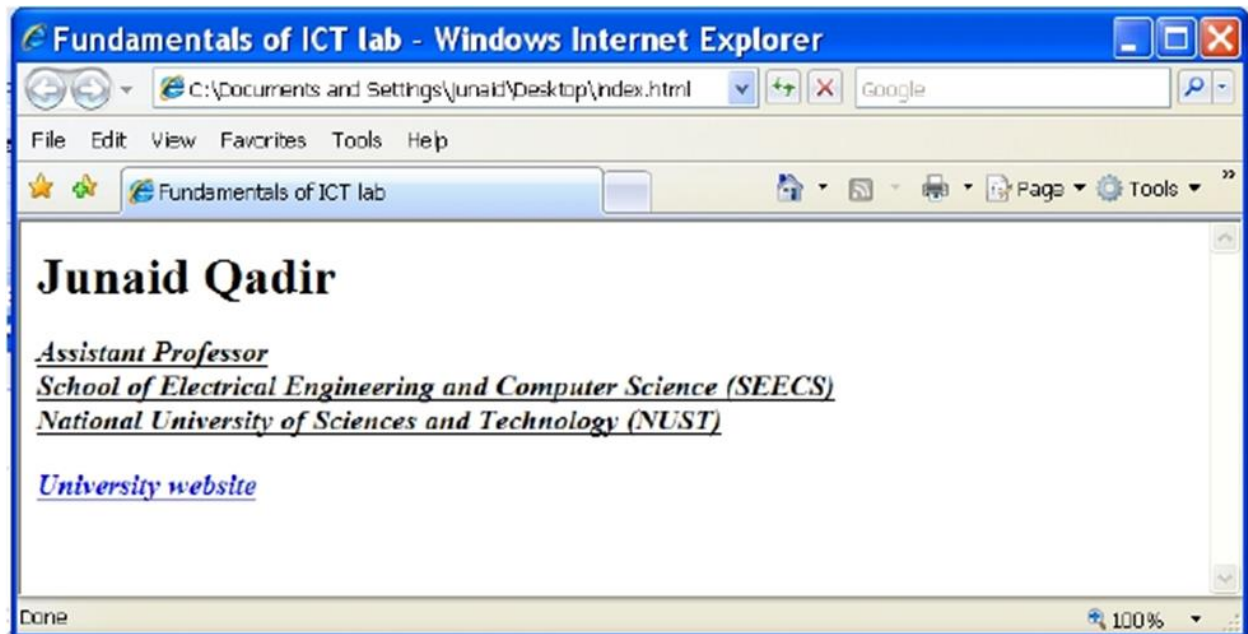
```
<html>
<head>
<title>Fundamentals of ICT lab</title>
</head>
<body>
<p><h1> Junaid Qadir </h1></p>
<p><h3><i>Assistant Professor <br>
School of Electrical Engineering and Computer Science (SEECs) <br>
National university of Sciences and Technology (NUST) <br></p>
<a href="http://www.niit.edu.pk/~junaid.qadir">University
website</a> </i></h3>
</body>
</html>
```



Underlining: Similar to the approach shown above, use the `<u>` and `</u>` tags for underlining some text. Open the index.html file on the Desktop and make the changes as highlighted in the figure below.



After making the changes above, save the file and then open it using a browser. You should see the browser display similar to as shown below.



If the browser display is similar to as shown above, save the index.html file (on the Desktop) to the ICTLab6 folder as **Underline.html**.

9. Inserting an image

To insert an image in an HTML page, `<img>` tag is used.

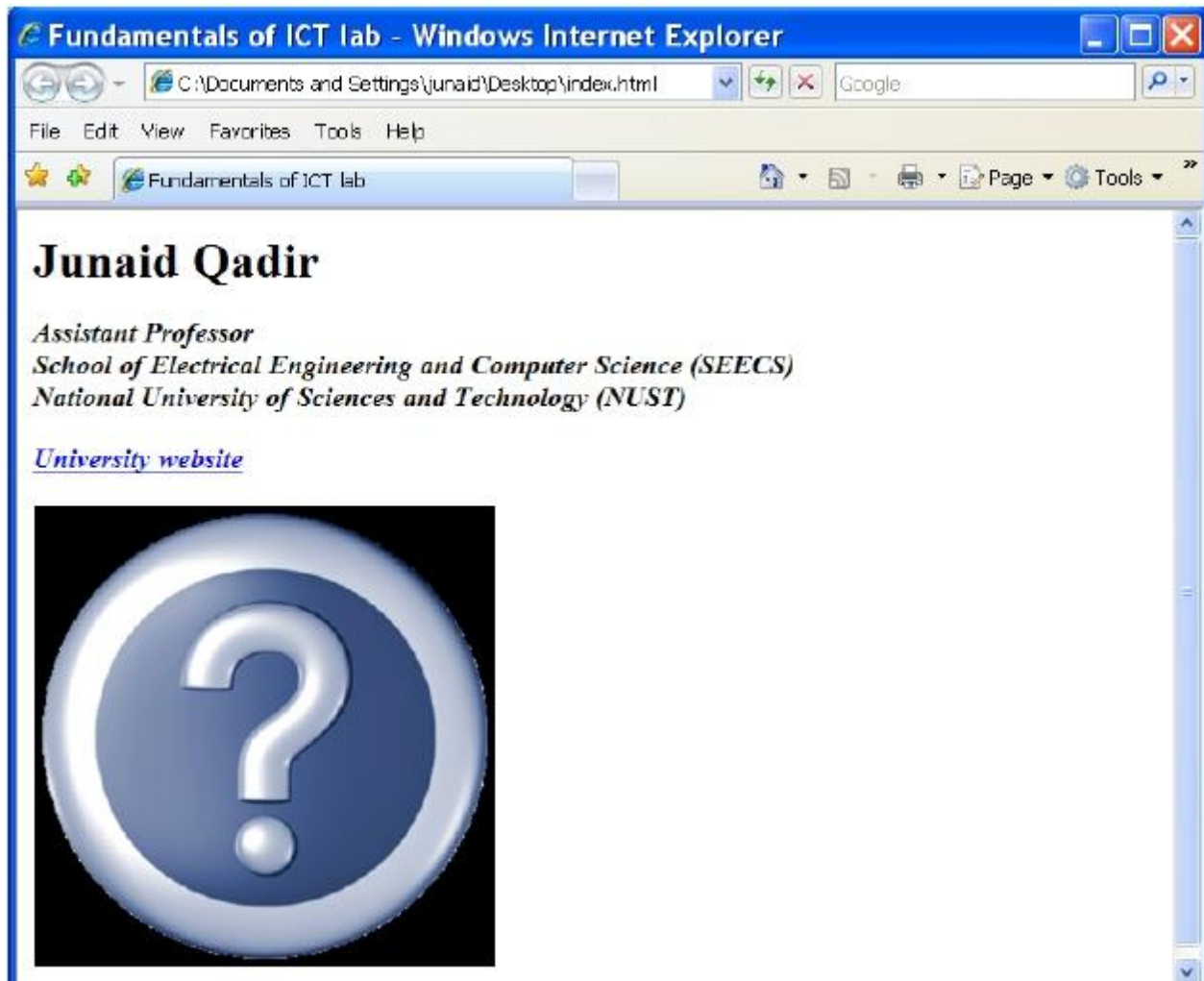
We will now demonstrate how an image is added to an HTML page. Modify the ndex.html page on the Desktop to as shown below and save the file.

```
index - Notepad
File Edit Format View Help
<html>
<head>
<title>Fundamentals of ICT lab</title>
</head>
<body>
<p><h1> Junaid Qadir </h1></p>
<p><h3><i>Assistant Professor <br>
School of Electrical Engineering and Computer Science (SEECs) <br>
National University of Sciences and Technology (NUST) <br></i></h3>
<a href="http://www.niit.edu.pk/~junaid.qadir">University
website</a> </i></h3>

</body>
```

The HTML page will include the file question-mark.png as an image. For this to work, question-mark.png should be in the same folder as the index.html file. Download the question-mark.png file from the course website and also place on the Desktop.

With the image file also on the Desktop, open the index.html file using a browser. You should be able to see the image as part of the web page displayed in the browser (as shown below).



If you find the browser display as shown above, copy the index.html page to ICTLab6 folder and rename it as **Image.html**. You **must** also copy the question-mark.png file to the ICTLab6 folder.

10. Employ style sheets

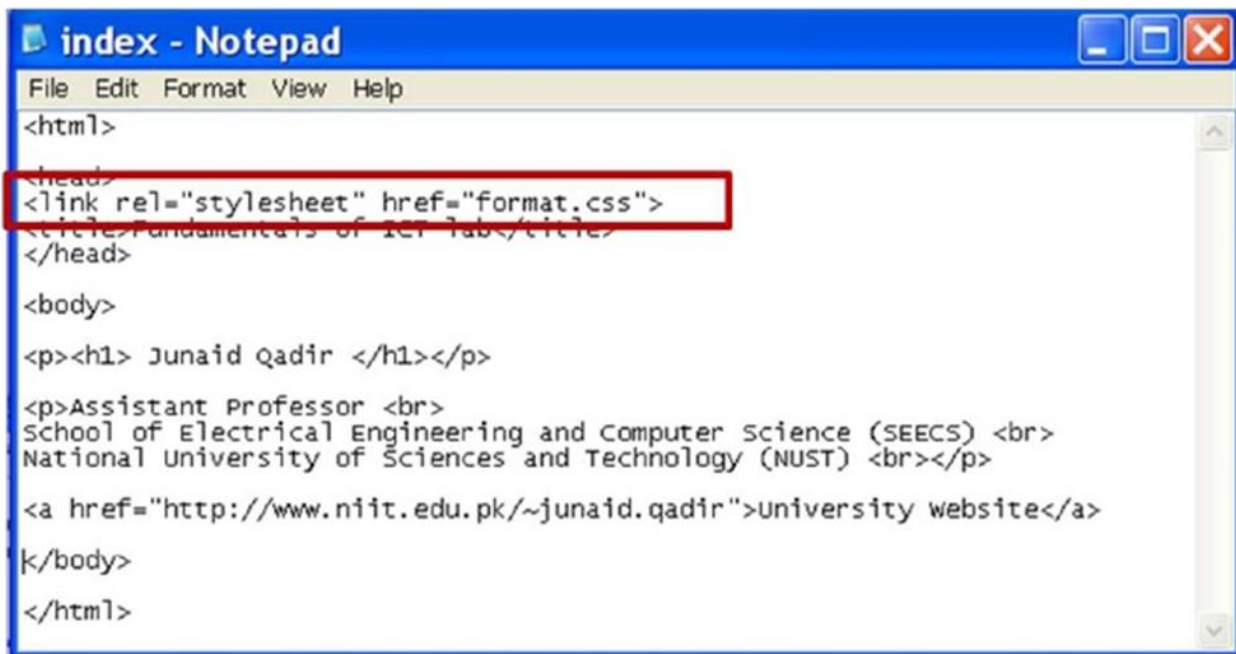
A Cascading Style Sheet (CSS) can be used to define the styling format of an HTML page. To use CSS in our HTML page, first of all, we will have to define our style in a file; secondly, that file must be linked to the HTML page. *[This should become clear soon.]* Open the **format.css** file on the Desktop and modify it to as shown in the figure below.



```
format - Notepad
File Edit Format View Help
h1 {
font-family: verdana,helvetica,sans serif;
font-weight: bold;
}
```

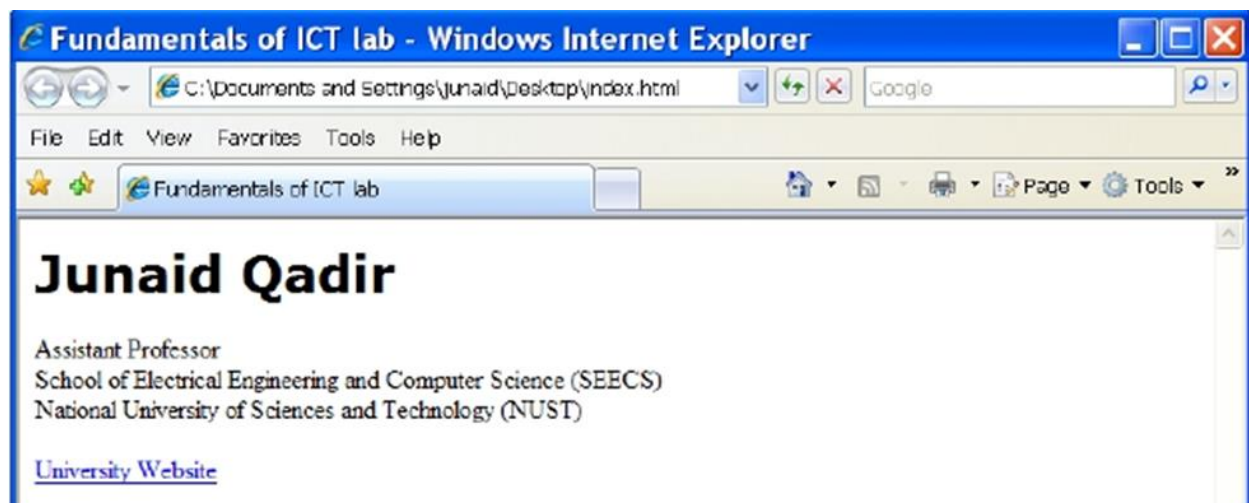
The style sheet above is defining how text marked with <h1> tag should be displayed in a browser.

After creating a style sheet, it has to be included in the web page too. This is done using the **<link>** tag as shown in the figure below.



```
index - Notepad
File Edit Format View Help
<html>
<head>
<link rel="stylesheet" href="format.css">
<title>Fundamentals of ICT lab</title>
</head>
<body>
<p><h1> Junaid Qadir </h1></p>
<p>Assistant Professor <br>
School of Electrical Engineering and Computer Science (SEECS) <br>
National University of Sciences and Technology (NUST) <br></p>
<a href="http://www.niit.edu.pk/~junaid.qadir">university website</a>
</body>
</html>
```

The display of the web page in a browser will now follow the style dictated by the `format.css` file.



The format defined by the CSS file can be further modified to affect all parts of the HTML page. As an example, we can also modify the default font size associated with heading types. This is shown below for the heading type **h1**.



Using the format.css file above, the browser display would look the following



If the browser display is as shown above, save the index.html and the format.css files (on the Desktop) in the ICTLab6 folder as **CSS.html** and the **Format.css**, respectively.

Part II---Basic Javascript (Dynamic web pages).

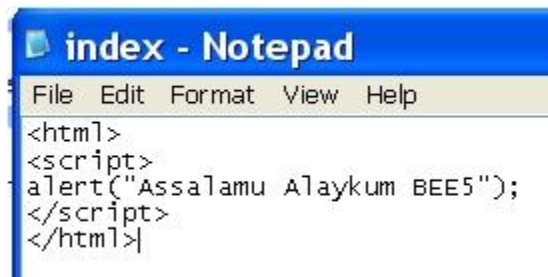
Static web pages (that use simple HTML) do not allow users to interact with the website. Dynamic web pages allow a user to interact with the website, and customize content according to the user.

The `<script>` `</script>` tag set specifies to the browser that any code between the tags is not written in HTML, but in a scripting language. You can place script tags anywhere in an HTML document. All of the code between `<script>` and `</script>` is written according to the rules of JavaScript. JavaScript is a different language than HTML, and its rules are different.

1. Output a message prompt

Create a new file on the Desktop and name it as Javascript.html.

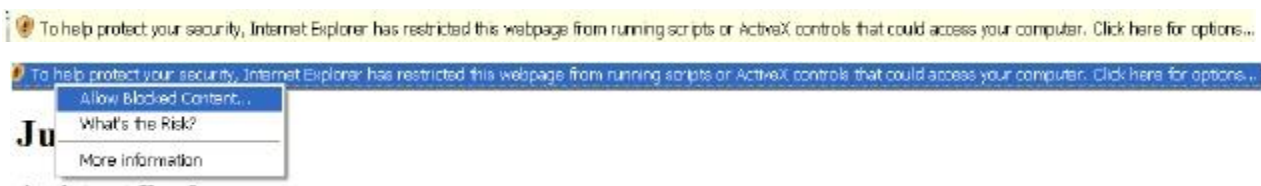
Modify the Javascript.html file to as shown in the figure below.



Save the file and open it using a browser. You should be able to find a message prompt similar to the prompt shown in the figure below.



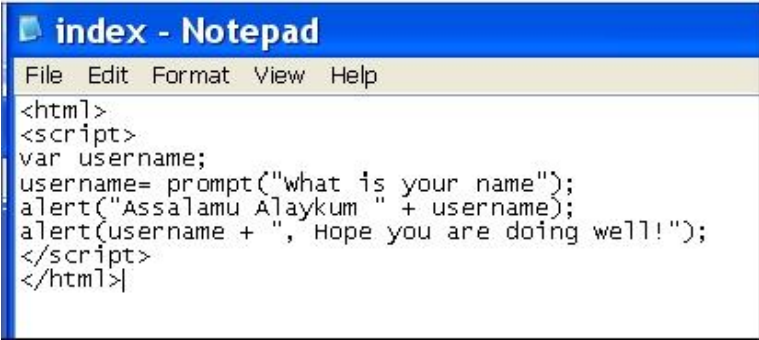
[Note: Internet Explorer users---If you see a warning prompt such as the following, allow the Blocked Content]



If you are receiving a prompt as shown above when the Javascript.html file is opened, save the **Javascript1.html** file in the ICTLab6 folder.

2. Prompt for user's name and then output a customized greeting.

Modify the Javascript.html file on the Desktop to as shown in the figure below.



```
index - Notepad
File Edit Format View Help
<html>
<script>
var username;
username= prompt("what is your name");
alert("Assalamu Alaykum " + username);
alert(username + ", Hope you are doing well!");
</script>
</html>
```

Save the file, and open it using a browser.

You will find that the browser will output a series of alerts (also called prompts) as shown below.



If you are receiving a prompt as shown above when the Javascript.html file is opened, save the **Javascript2.html** file in the ICTLab6 folder.

ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

W

LESSON 13: Computer Systems (Organization and Architecture)

OVERVIEW:

LEARNING OUTCOME:

At the end of the chapter, the students are able to be familiar with the structure and function of the computer system.

MATERIALS NEEDED:

Computer, Projector, Presentation Slides

DURATION:

240 minutes

LESSON PROPER:

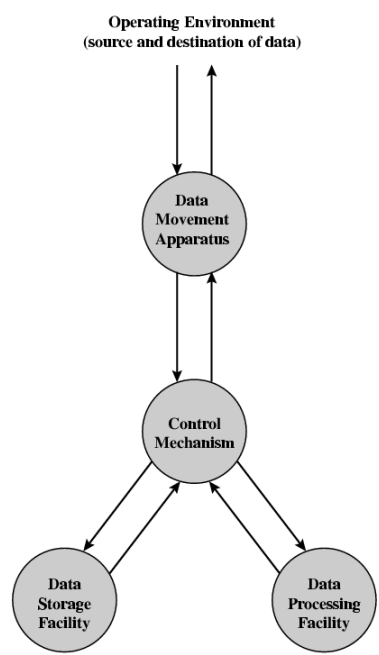
Architecture & Organization

- Architecture is those attributes visible to the programmer
 - Instruction set, number of bits used for data representation, I/O mechanisms, addressing techniques.
 - e.g. Is there a multiply instruction?
- Organization is how features are implemented
 - Control signals, interfaces, memory technology.
 - e.g. Is there a hardware multiply unit or is it done by repeated addition?
- All Intel x86 family share the same basic architecture
- The IBM System/370 family share the same basic architecture
- This gives code compatibility
 - At least backwards
- Organization differs between different versions

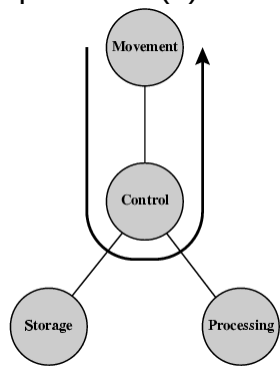
Function

- All computer functions are:
 - Data processing
 - Data storage
 - Data movement
 - Control

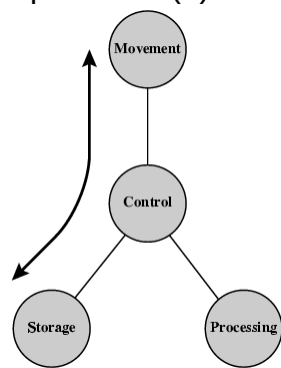
Functional View



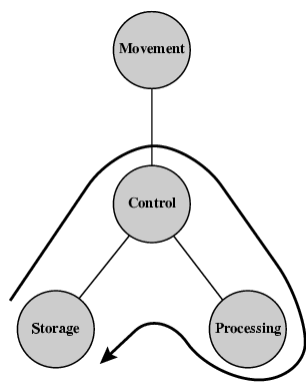
Operations (a) Data movement



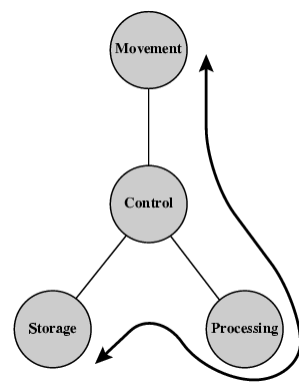
Operations (b) Storage



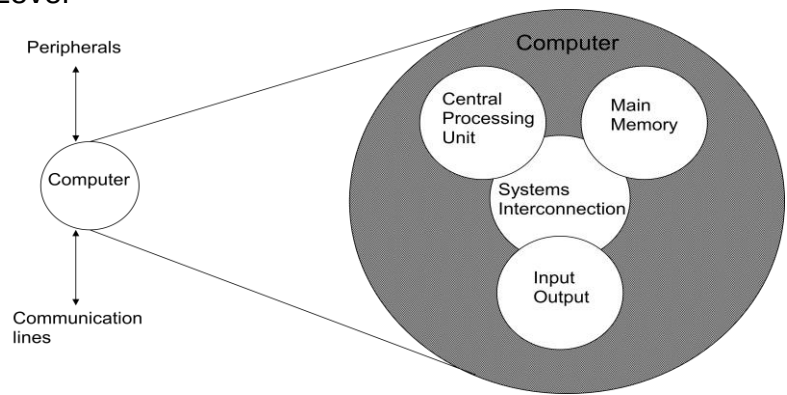
Operation (c) Processing from/to storage to I/O



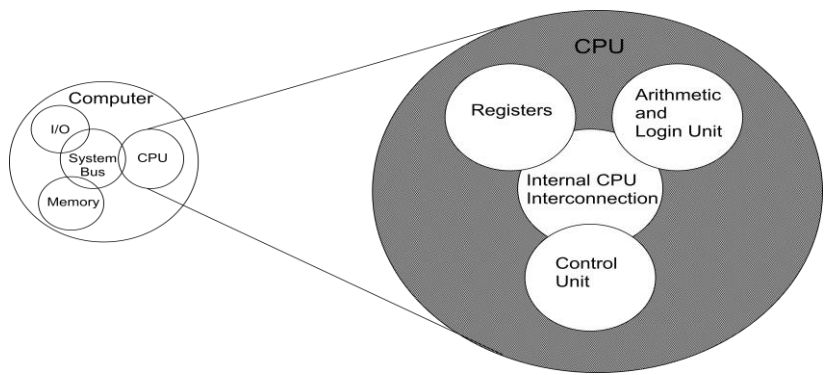
Operation (d) Processing from storage



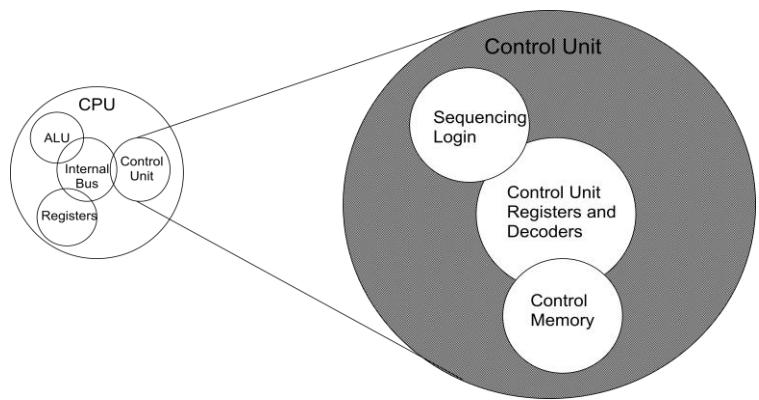
Structure - Top Level



Structure - The CPU



Structure - The Control Unit



ACTIVITY:

Oral Presentation, Group Discussion

EVALUATION:

W